Instead of directly working out the condition in the manner indicated above, I present the investigation in a synthetic form as follows: Taking v, v' for the vector parts of the two given quaternions, so that $q = w + v$, $q' = w' + v'$, I write for shortness

$$\begin{aligned}
\theta &= w - w', \\
\alpha &= v^2 + v'^2, \quad = -x^2 - y^2 - z^2 - x'^2 - y'^2 - z'^2, \\
\beta &= v^2 - v'^2, \quad = -x^2 - y^2 - z^2 + x'^2 + y'^2 + z'^2, \\
D &= -\theta(\alpha - \theta^2), \\
A &= \beta - \theta^2, \\
B &= \beta + \theta^2,
\end{aligned}$$

so that θ, α, β, D, A, B are all of them scalars. With these I form a quaternion $Q = (D + Av)(D + Bv')$; I say that we have identically

$$qQ - Qq' = \{D - v \cdot v'^2 + v' \cdot v^2 + w \cdot \theta\}(\theta^4 - 2\alpha\theta^2 + \beta^2).$$

It of course follows that, if $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$, then $qQ - Qq' = 0$, viz. $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$ is the condition $\Omega = 0$, for the existence of the required quaternion; and this condition being satisfied, then (omitting the arbitrary scalar factor) the value of the quaternion is $Q = (D+Av)(D+Bv')$, a value giving $T(Q) = 0$, that is, $W^2+X^2+Y^2+Z^2 = 0$. If $\theta = 0$ (that is, $w = w'$), then the condition becomes $\beta = 0$, that is, $x^2+y^2+z^2-x'^2-y'^2-z'^2 = 0$; and these two conditions being satisfied, Q ceases to have the determinate value given by the foregoing formula: it has a value involving an arbitrary parameter, and is no longer such that $W^2 + X^2 + Y^2 + Z^2 = 0$.

The identical equation is at once verified; we have

$$\begin{aligned}
qQ - Qq' &= (w + v)Q - Q(w' + v') \\
&= \theta \cdot (D^2 + DAv + DBv' + ABvv') \\
&\quad + v \cdot (D^2 + DAv + DBv' + ABvv') \\
&\quad - (D^2 + DAv + DBv' + ABvv')v' \\
&= \theta D^2 + DAv^2 - DBv'^2 \\
&\quad + v(DA\theta + D^2 - ABv'^2) \\
&\quad + v'(DB\theta + ABv^2 - D^2) \\
&\quad + vv'(\theta AB + DB - DA).
\end{aligned}$$

The first line is here $= D \cdot \{D\theta + Av^2 + Bv'^2\}$, viz. the term in $\{\ \}$ is

$$-(\alpha - \theta^2)\theta + (\beta - \theta^2)v^2 - (\beta + \theta^2)v'^2, \quad = -\alpha\theta^2 = \theta\cdot;$$

and similarly each of the other lines contains the factor $\theta^4 - 2\alpha\theta^2 + \beta^2$, and the equation is thus seen to hold good.

The tensor of Q is $= (D^2 - A^2v^2)(D^2 - B^2v'^2)$; and we have

$$\begin{aligned} D^2 - A^2v^2 &= \theta^2(\alpha - \theta^2)^2 - (\beta - \theta^2)^2v^2 \\ &= \theta^6 - (2\alpha + v^2)\theta^4 + (\alpha^2 + 2\beta v^2)\theta^2 - \beta^2v^2, \end{aligned}$$

which, observing that $\alpha^2 + 2\beta v^2 = \beta^2 + 2v^2(2v^2 - \alpha)$ is $= (\theta^2 - v'^2)^2$ and similarly $D^2 - B^2v'^2$ is $= (\theta^2 - v^2)^2$; hence the tensor is

$$TQ = (\theta^2 - v^2)(\theta^2 - v'^2),$$

which, in virtue of $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$, is $= 0$.

The particular case is when $Tq - Tq' = 0$, that is, $w^2 - v^2 - w'^2 + v'^2 = 0$, or say $w^2 - w'^2 = v^2 - v'^2$, that is, $\theta(w + w') = \beta$. Combining with this the general condition $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$, we find $\theta^2\{\theta^2 - 2\alpha + (w + w')^2\} = 0$, or in the second factor, for θ and α substituting their values, we have $2\theta^2(w^2 - v^2 + w'^2 - v'^2) = 0$, that is, $2\theta^2(Tq + Tq') = 0$. Attending to the assumed relation $Tq - Tq' = 0$, the second factor can only vanish if $Tq = 0$, $Tq' = 0$; hence, disregarding this more special case, the factor which vanishes must be the first factor, that is, $\theta = 0$; or the equations $Tq - Tq' = 0$ and $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$ give $\beta = 0$ and $\theta = 0$, that is, we have as already mentioned $w - w' = 0$, and $x^2 + y^2 + z^2 = x'^2 + y'^2 + z'^2$, viz. the given quaternions have their scalars equal, and the squares of their vectors also equal. The equation here is $vQ - Qv' = 0$; and writing $v^2 = v'^2 = -\rho^2$, we see at once that a solution is

$$Q = (-\rho^2 + vv') + \lambda(v + v'),$$

where λ is an arbitrary scalar; in fact, with this value of Q, we have at once vQ and Qv' each $= \rho^2(v + v') + \lambda(\rho^2 + vv')$; and the equation $vQ - Qv' = 0$ is thus satisfied.

The value of the tensor is easily found to be

$$TQ = 2(\lambda^2 + \rho^2)(\rho^2 + xx' + yy' + zz'),$$

which is not $= 0$.

In accordance with a remark in the introductory paragraphs, the solution $Q = -\rho^2 + vv' + \lambda(v + v')$ is not comprised in the general solution. As to this, observe that, in the case in question $\theta = 0$, $\beta = 0$, we have from the general theorem the form that is,

$$Q = -\rho^2\theta^2(\theta^2 - 2v'^2) \pm \theta^2(\theta^2 - 2v^2)v', \quad = 0.$$

In the condition $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$, writing $\theta = 0$, we have $\theta^2 = 2\alpha$, or for α writing its value $-2\rho^2$, we have $\theta^2 = -4\rho^2$, $\frac{\theta^2}{4} = -\rho^2$, and thence $\frac{A}{B} = \rho^2$, and $\frac{A}{B} = \lambda\rho$, if λ denote the $\sqrt{(-1)}$ of ordinary algebra. The resulting formula is thus $Q = -\rho^2 + vv' \pm \lambda\rho(v + v')$, which corresponds to the determinate value $\pm\lambda\rho$ of the constant λ.

in the condition $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$, writing $\theta = 0$, we have $\frac{\theta^2}{4} = 2\alpha$, or for α writing its value $= 2\rho^2$, we have $\theta^2 = -4\rho^2$, $\frac{A^2}{B^2} = -\rho^2$, and thence $\frac{A^2}{B^2} = \rho^2$, $\frac{A}{B} = \lambda\rho$, if λ denote the $\sqrt{(-1)}$ of ordinary algebra. The resulting formula is thus

$$Q = -\rho^2 + vv' \pm \lambda\rho(v + v'),$$

which corresponds to the determinate value $\pm\lambda\rho$ of the constant λ.

The foregoing solution $Q = -\rho^2 + vv' + \lambda(v + v')$ may be easily identified with that given pp. 124, 125 of Tait's *Elementary Treatise on Quaternions*; the case there considered is that of a real quaternion, and it was therefore assumed that the two conditions $w = w'$, and $x^2 + y^2 + z^2 = x'^2 + y'^2 + z'^2$, were each of them satisfied.

The theory of quaternions is, as is well known, identical with that of matrices of the second order; the identity is, in effect, established by the remark and footnote "Linear Associative Algebra," *American Journal of Mathematics*, t. IV. (1881), p. 132. Writing x, y, z, w for Peirce's imaginaries i, j, k, 1, these have the multiplication table:

	x	y	z	w
x	x	y	0	0
y	0	0	x	y
z	z	w	0	0
w	0	0	z	w

Thus if λ, $= \sqrt{(-1)}$, be the imaginary of ordinary algebra, and i, j, k, the quaternion imaginaries, the relations between i, j, k and x, y, z, w are

$$\begin{aligned} x &= \tfrac{1}{2}(1 - \lambda i), &\quad \text{or conversely}\quad 1 &= x + w, \\ y &= \tfrac{1}{2}(j - \lambda k), & i &= \lambda(x - w), \\ z &= \tfrac{1}{2}(-j - \lambda k), & j &= (y - z), \\ w &= \tfrac{1}{2}(1 + \lambda i), & k &= \lambda(y + z), \end{aligned}$$

and we can thus at once express a quaternion as a linear function of the x, y, z, w, or a linear function of the x, y, z, w as a quaternion. And we then consider

$ax + by + cz + dw$ as denoting the matrix $\begin{vmatrix} a, b \\ c, d \end{vmatrix}$; we obtain for the product of two matrices the ordinary formula

$$\begin{vmatrix} a, b \\ c, d \end{vmatrix} \begin{vmatrix} a', b' \\ c', d' \end{vmatrix} = \begin{vmatrix} (a, b)(a', c'), & (a, b)(b', d') \\ (c, d)(a', c'), & (c, d)(b', d') \end{vmatrix},$$

viz. we have

$$\begin{aligned} (ax + by + cz + dw)(a'x + b'y + c'z + d'w) &= aa'x^2 + bc'yz + \&\text{c.}, \\ &= (aa' + bc')x + (ab' + bd')y + (ca' + dc')z + (cb' + dd')w, \end{aligned}$$

in accordance with the formula for the product of the two matrices. Observe that, writing

$$A + Bi + Cj + Dk = A(x + w) + B\lambda(x - w) + C(y - z) + D\lambda(y + z) = ax + by + cz + dw,$$

we have

$$\begin{vmatrix} a, b \\ c, d \end{vmatrix} = \begin{vmatrix} A + B\lambda, & -C + D\lambda \\ C + D\lambda, & A - B\lambda \end{vmatrix},$$

and thence $ad - bc = A^2 + B^2 + C^2 + D^2$, so that the determinant of the matrix corresponds to the tensor of the quaternion.

837.

ON THE SO-CALLED D'ALEMBERT–CARNOT GEOMETRICAL PARADOX.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 113, 114.]

THE present note has reference to Prof. Sylvester's paper on this subject [*l. c.*, pp. 92–96]. I cannot admit that "D'Alembert and Carnot raised a well-founded objection" to the then and even now too prevalent interpretation of the meaning of the geometrical positive and negative: it appears to me that the objection was not a well-founded one.

Consider through the origin K an indefinite line $t'Kt$, and measure off from K in the sense Kt a distance equal to the positive quantity a, and let m be the extremity of the distance thus measured off. There is not in the ordinary theory any reason why the distance Km should be $= +a$ rather than $= -a$; it is $= +a$, if Kt be the positive sense of the line through K, and it is $= -a$ if Kt' be the positive sense of the line through K; if it be undetermined which of the two is the positive sense, then the distance Km is $= \pm a$, the sign being essentially indeterminate.

The problem is from a point K outside a given circle to draw a line Kmm' such that the intercepted portion mm' within the circle has a given value c. Supposing that the line from K to the centre meets the circle in the points A, B at the distances $KA = a$, $KB = b$: then if $Km = r$, we have $ab = r(c+r)$, or $r = -\frac{1}{2}c \pm \sqrt{(\frac{1}{4}c^2 + ab)}$; viz. we have for r, not simultaneously but alternatively, the positive value $-\frac{1}{2}c + \sqrt{(\frac{1}{4}c^2 + ab)}$, and the negative value $-\frac{1}{2}c - \sqrt{(\frac{1}{4}c^2 + ab)}$, the latter of these being the greatest in absolute magnitude; say the values are $+\rho_1$ and $-\rho_2$. We may with either of these values construct the point m; viz. we obtain m as one of the intersections of the given circle with the circle centre K and radius ρ_1, or else with the circle centre K and radius ρ_2 (that is, radius ρ_2); and attending to the intersections on the same side of the line from K to the centre, it happens that the two points m thus determined are on one and the same line $t'Kt$; but there is no *a priori* reason why the positive senses should be the same, and they are in fact opposite to each other, in the two cases respectively; in the one case we measure off the distance ρ_1 in the sense Kt, in the other case the distance $-\rho_2$ in the sense Kt'; that is, we in fact measure off the positive distances $+\rho_1$, and $+\rho_2$, in one and the same sense Kt; thus obtaining for the point m one or the other extremity of a determinate secant through K.

The best illustration is I think in the elementary problem of finding the perpendicular distance of a given line from the origin. Let $Ax + By + C = 0$ be the equation of the given line: and first let a line be drawn in a determinate sense, say at the inclination θ to the positive part of the axis of x, to meet the given line. Taking r for the distance from the origin of the point of intersection, we have, for the coordinates of the point of intersection, x, $y = r\cos\theta$, $r\sin\theta$; and thence $r(A\cos\theta + B\sin\theta) + C = 0$, that is,

$$r = \frac{-C}{A\cos\theta + B\sin\theta},$$

a perfectly determinate value. But the perpendicular on the given line may be considered as drawn in one or the other of two opposite senses; that is, we have at pleasure

$$\cos\theta, \sin\theta = \frac{A}{\sqrt{(A^2+B^2)}}, \frac{B}{\sqrt{(A^2+B^2)}}, \quad \text{or else} \quad \frac{-A}{\sqrt{(A^2+B^2)}}, \frac{-B}{\sqrt{(A^2+B^2)}},$$

and thence

$$r = \frac{-C}{\sqrt{(A^2+B^2)}}, \quad \text{or else} \quad r = \frac{C}{\sqrt{(A^2+B^2)}};$$

that is, the perpendicular distance is $= \frac{\pm C}{\sqrt{(A^2+B^2)}}$, with the essentially indeterminate sign $\pm$, because the distance may be considered as drawn from the origin in one or the other of the two opposite senses.

838.

ON THE TWISTED CUBICS UPON A QUADRIC SURFACE.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 129–132.]

A CUBIC (twisted cubic) on a quadric surface meets each generator of the one kind twice, and each generator of the other kind once (Salmon, *Solid Geometry*, Ed. 4, p. 301). There are thus on the surface two kinds of cubics, viz. distinguishing for convenience the generating lines as generators and directors, a cubic may meet each generator twice and each director once; or it may meet each generator once and each director twice. And two cubic curves are accordingly of the same kind or of different kinds.

Consider for example the quadric surface $xw - yz = 0$, and the cubic

$$x : y : z : w = 1 : \phi : \phi^2 : \phi^3.$$

If we call the line $(x = ky, kw = z)$ a generator, and the line $(x = kz, kw = y)$ a director, for the intersections with a generator we have $1 = k\phi$, $k\phi^3 = \phi^2$; equations with the common root $k\phi = 1$, or there is a single intersection; for the intersections with a director, $1 = k\phi^2$, $k\phi^3 = \phi$, equations with the common roots $k\phi^2 = 1$, or there are two intersections.

Consider on the same quadric surface the cubic

$$x : y : z : w = \overline{\theta - \alpha}\cdot\overline{\theta - \beta}\cdot\overline{\theta - \gamma} : \overline{\theta - \alpha}\cdot\overline{\theta - \epsilon}\cdot\overline{\theta - \zeta} : \overline{\theta - \delta}\cdot\overline{\theta - \beta}\cdot\overline{\theta - \gamma} : \overline{\theta - \delta}\cdot\overline{\theta - \epsilon}\cdot\overline{\theta - \zeta};$$

or as, for shortness, I write these,

$$x : y : z : w = \alpha\beta\gamma : \alpha\epsilon\zeta : \delta\beta\gamma : \delta\epsilon\zeta,$$

viz. α is written to denote $\theta - \alpha$, and so in other cases; for the intersections with a generator, we have $\alpha\beta\gamma = k\alpha\epsilon\zeta$, $k\delta\epsilon\zeta = \delta\beta\gamma$, equations having the two common roots

$\beta\gamma = k\epsilon\zeta$; or there are two intersections with the generator. And in like manner there is one intersection with the director. The cubic

$$x : y : z : w = \alpha\beta\gamma : \alpha\epsilon\zeta : \delta\beta\gamma : \delta\epsilon\zeta$$

is thus a cubic of different kind from

$$x : y : z : w = 1 : \phi : \phi^2 : \phi^3.$$

And in the same manner it appears that the cubic

$$x : y : z : w = \alpha\beta\gamma : \delta\beta\gamma : \alpha\epsilon\zeta : \delta\epsilon\zeta$$

is of the same kind with

$$x : y : z : w = 1 : \phi : \phi^2 : \phi^3.$$

The two cubics of different kinds intersect in 5 points; the two cubics of the same kind in only 4 points. In fact, for the intersections with the cubic

$$x : y : z : w = 1 : \phi : \phi^2 : \phi^3,$$

we have $xz - y^2 = 0$, $yw - z^2 = 0$, and substituting herein the θ-values, $x : y : z : w = \alpha\beta\gamma : \alpha\epsilon\zeta : \delta\beta\gamma : \delta\epsilon\zeta$, we find

$$\alpha\delta\beta^2\gamma^2 - \alpha^2\epsilon^2\zeta^2 = 0, \quad \delta\epsilon\zeta\alpha\epsilon\zeta - \delta^2\beta^2\gamma^2 = 0,$$

equations with the common five roots $\delta\beta^2\gamma^2 - \alpha\epsilon^2\zeta^2 = 0$; whereas for the θ-values $x : y : z : w = \alpha\beta\gamma : \delta\beta\gamma : \alpha\epsilon\zeta : \delta\epsilon\zeta$, we have

$$\alpha^2\beta\gamma\epsilon\zeta - \delta^2\beta^2\gamma^2 = 0, \quad \delta^2\beta\gamma\epsilon\zeta - \alpha^2\epsilon^2\zeta^2 = 0,$$

equations with the common four roots $\alpha^2\epsilon\zeta - \delta^2\beta\gamma = 0$.

I remark in passing that the equations just obtained have each of them a root $\theta = \infty$, but this would have been avoided if the θ-equations had been taken in the slightly altered form

$$x : y : z : w = \alpha\beta\gamma : b\alpha\epsilon\zeta : c\delta\beta\gamma : bc\delta\epsilon\zeta, \text{ and } x : y : z : w = \alpha\beta\gamma : c\delta\beta\gamma : b\alpha\epsilon\zeta : bc\delta\epsilon\zeta,$$

b, c being arbitrary constants; and the special value is thus of no importance.

The two cubics intersecting in 5 points constitute the complete intersection of a quadric surface and a cubic surface; and conversely when a quadric surface and a cubic surface intersect in two cubics, then the cubics must have 5 intersections, and are thus by what precedes cubics of different kinds in regard to the quadric surface. In fact, considering two cubics meeting in 5 points, we can, through these 5 points, through 2 points assumed at pleasure on the first cubic, and through 2 points assumed at pleasure on the second cubic, draw a determinate quadric surface: this meets each of the two cubics in $5+2$ points, and consequently it entirely contains the cubic; viz. we have through the two cubics a determinate quadric surface. Again, through the 5 points, through 5 points at pleasure on the first cubic and through 5 points at pleasure on the second cubic, we may draw a cubic surface; this meets each cubic in $5+5$ points, and therefore it entirely contains the cubic; we have thus a cubic surface through the two cubics. The cubic surface has been subjected only to $5+5+5=15$ conditions, and the equation thus contains homogeneously $20-15, =5$ constants; viz. if $\Omega=0$ be the quadric surface through the two cubics the general form is

$$(\alpha x+\beta y+\gamma z+\delta w)\Omega+\lambda U=0;$$

disregarding the term in Ω, we have thus, in fact, a determinate cubic $U=0$.

Taking as above the first cubic as given by the equations $x:y:z:w=1:\phi:\phi^2:\phi^3$, and the second cubic as given by the equations $x:y:z:w=\alpha\beta\gamma:\alpha\epsilon\zeta:\delta\beta\gamma:\delta\epsilon\zeta$, the equation of the cubic surface which contains the two cubics will be of the form

$$(Ax+By+Cz+Dw)(yw-z^2)+(A'x+B'y+C'z+D'w)(xz-y^2)=0.$$

Substituting herein the θ-values and omitting the common factor $\alpha\epsilon^2\zeta^2-\delta\beta^2\gamma^2$, we have

$$\delta(A\alpha\beta\gamma+B\alpha\epsilon\zeta+C\delta\beta\gamma+D\delta\epsilon\zeta)-\alpha(A'\alpha\beta\gamma+B'\alpha\epsilon\zeta+C'\delta\beta\gamma+D'\delta\epsilon\zeta)=0,$$

which is of the fourth order in θ. We may assume $C'=0$, and $D'=0$; in fact, the terms in C' and D' might be got rid of by the substitutions $z(xz-y^2)=y(xw-yz)-x(yw-z^2)$, $w(xz-y^2)=z(xw-yz)-y(yw-z^2)$; we then have 6 coefficients A, B, C, D, A', B', and equating to zero the terms in θ^0, θ^1, θ^2, θ^3, θ^4, the ratios of these are determined by the five equations: and we have thus the required cubic surface.

Starting from either cubic considered as a curve on the given quadric surface; if through the centre we describe a cubic surface, this meets the quadric surface in a second cubic and the two cubics intersect each other in 5 points. In particular, if the one cubic is $x:y:z:w=1:\phi:\phi^2:\phi^3$, it appears by what precedes that the other cubic may be taken to be $x:y:z:w=\alpha\beta\gamma:\alpha\epsilon\zeta:\delta\beta\gamma:\delta\epsilon\zeta$, intersecting the former in the 5 points given by the equation $\delta\beta^2\gamma^2-\alpha\epsilon^2\zeta^2=0$. The five points are of course points of contact of the quadric surface and the cubic surface.

A very simple instance is the following. The quadric surface $xw-yz=0$ and the cubic surface

$$y(xz-y^2)+z(yw-z^2)=0$$

intersect in the two cubics

$$x:y:z:w=1:\phi:\phi^2:\phi^3 \quad \text{and} \quad x:y:z:w=1:\theta^2:\theta:\theta^3.$$

The two cubics meet in the 5 points

$$\theta = (0, \infty, 1, \omega, \omega^2); \quad \phi = (0, \infty, 1, \omega^2, \omega),$$

or, what is the same thing, in the 5 points

$$(x, y, z : w) = (1, 0, 0, 0), (0, 0, 0, 1), (1, 1, 1, 1), (1, \omega, \omega^2, 1), (1, \omega^2, \omega, 1),$$

where ω is an imaginary cube root of unity.

I was anxious to work out this result not for its own sake, but as an illustration of a point which first arises as to octics: a unicursal octic curve is not in general situate on any surface lower than a quartic surface; if on the octic we take at pleasure 33 points, we may through these draw two quartic surfaces $U = 0$, $V = 0$, each of which will entirely contain the curve: the two surfaces meet in a second octic also unicursal, and the two curves intersect in 34 points, which are points of contact of the two quartic surfaces. We cannot, by adjoining to the given octic any curve of an inferior to its own, obtain a complete intersection of two surfaces: the most simple case is when we adjoin to the given octic as above another unicursal octic, and thus obtain a complete intersection of two quartic surfaces. It is to be observed that the second curve is determined completely and uniquely by the given curve: considering the former curve as given by the expressions of the coordinates in terms of a parameter ϕ, the determination of the like expressions for the latter curve would appear to be a problem of very great difficulty.

839.

ON THE MATRICAL EQUATION $qQ - Qq = 0$.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 176–178.]

I CONSIDER the matrical equation $qQ - Qq' = 0$, where q, q' are given matrices $\begin{vmatrix} a, b \\ c, d \end{vmatrix}$, $\begin{vmatrix} a', b' \\ c', d' \end{vmatrix}$, and Q, $= \begin{vmatrix} A, B \\ C, D \end{vmatrix}$, is a matrix which has to be determined. As remarked in the paper "On the Quaternion Equation $qQ - Qq' = 0$," *Messenger*, t. XIV. (1885), pp. 108–112, [836] the question for matrices is equivalent to that for quaternions: for a matrix $\begin{vmatrix} a, b \\ c, d \end{vmatrix}$ may be regarded as a quaternion

$$\tfrac{1}{2}(a+d) - \tfrac{1}{2}\lambda(a-d)i + \tfrac{1}{2}(b-c)j - \tfrac{1}{2}\lambda(b+c)k,$$

or (omitting the factor $\frac{1}{2}$) as the quaternion

$$(a+d) - \lambda(a-d)i + (b-c)j - \lambda(b+c)k,$$

where λ, $= \sqrt{(-1)}$, is the imaginary of ordinary algebra. Hence considering q, q' as denoting the quaternions

$$\begin{gathered}(a+d) - \lambda(a-d)i + (b-c)j - \lambda(b+c)k, \\ (a'+d') - \lambda(a'-d')i + (b'-c')j - \lambda(b'+c')k,\end{gathered}$$

we can, if a certain condition is satisfied, find a quaternion Q such that $qQ - Qq' = 0$; say this is $Q = W + iX + jY + kZ$; putting this we find

$$Q = \begin{vmatrix} A, B \\ C, D \end{vmatrix} = \begin{vmatrix} W + \lambda X, & Y - \lambda Z \\ -Y - \lambda Z, & W - \lambda X \end{vmatrix},$$

for the required matrix Q; this being an indeterminate matrix, such that $AD - BC = 0$.

But it is better to solve directly the matrical equation

$$\begin{vmatrix} (A,B)(a,b), & (A,B)(b',d') \\ (C,D)(a,b), & (C,D)(b',d') \end{vmatrix} - \begin{vmatrix} (a,b)(A,C), & (a,b)(B,D) \\ (c,d)(A,C), & (c,d)(B,D) \end{vmatrix} = 0,$$

viz.

$$\begin{vmatrix} (A,B)(a,b), & (A,B)(b',d') \\ (C,D)(a,b), & (C,D)(b',d') \end{vmatrix} - \begin{vmatrix} (a,b)(A,C), & (a,b)(B,D) \\ (c,d)(A,C), & (c,d)(B,D) \end{vmatrix} = 0,$$

that is,

$$\begin{aligned} Aa + Bc' &= aA + bC, \\ Ba + Dc' &= aB + bD, \\ Ac + Cd' &= cA + dC, \\ Bc + Dd' &= cB + dD, \end{aligned}$$

or, what is the same thing,

$$\begin{vmatrix} a-a', & -c', & 0, & b \\ -b', & a-d', & b, & 0 \\ c, & 0, & d-a', & -c' \\ 0, & c, & -b', & d-d' \end{vmatrix} (A,B,C,D) = 0,$$

viz. we have (A, B, C, D) connected by these four linear equations: the necessary condition is that the determinant formed out of the matrix which here presents itself shall be $= 0$.

After some reduction, and putting for shortness

$$\nabla = ad - bc, \quad \nabla' = a'd' - b'c',$$

this is found to be

$$(\nabla - \nabla')^2 + [\nabla(a' + d') - \nabla'(a + d)](a + d - a' - d') = 0;$$

which is the condition for the existence of a solution. This condition being satisfied, the four equations will be equivalent to three independent equations, which serve to determine A, B, C, D; and, assuming the absolute value of A, we find

$$\begin{aligned} A &= -(a' + d' - a - d), \\ \nabla B &= \nabla' \cdot b' + \nabla(a' + d' - a - d) + ab' - a'b, \\ \nabla C &= \nabla' \cdot c' + \nabla(a' + d' - a - d) + dc' - d'c, \\ \nabla D &= D \cdot a' - (d' - d)(a' + d' - a - d) - cc'(a' + d' - a - d) + b'c(a' + d' - a - d)'; \end{aligned}$$

values which give

$$ad' + b'c = ad - bc;$$

viz. in the case where the given matrices are such that $ad + bg = (a' + d') - bc - bc' + ad'$, the components A, B, C, D of the required matrix Q are determined as such that $AD - BC = 0$.

If we have $\nabla - \nabla' = 0$, that is, $ad - bc = a'd' - b'c'$, then the condition becomes $a' + d' = a + d$; and these two conditions being satisfied, the four equations reduce themselves to two independent equations, and the ratios $A : B : C : D$ have no longer determinate values; the four equations may in fact be written

$$\begin{vmatrix} 0, & b, & -c', & a - a' \\ -b, & 0, & a + d', & b' \\ c, & a' - d, & 0, & -c' \\ a - d, & c, & -b, & 0 \end{vmatrix} (D, C, B, A) = 0,$$

which are of the form
$$\begin{vmatrix} 0, & -h, & g, & a \\ h, & 0, & -f, & b \\ -g, & f, & 0, & c \\ -a, & -b, & -c, & 0 \end{vmatrix},$$

where the coefficients a, b, c, f, g, h are such that $af + bg + ch = 0$; and the four equations are thus equivalent to two independent equations.

To obtain a symmetrical solution, assume a relation with arbitrary coefficients ξ, η, ζ, ω; we then find

$$\begin{aligned} D &= -c\eta - h\zeta + b\omega, \\ B &= h\eta + f\zeta + a\omega, \\ C &= -g\eta + a\xi + g\omega, \\ A &= -b\xi - f\eta + h\omega, \end{aligned}$$

viz. the complete theorem is that, if the matrices

$$q = \begin{vmatrix} a, b \\ c, d \end{vmatrix}, \quad q' = \begin{vmatrix} a', b' \\ c', d' \end{vmatrix},$$

are such that $ad - bc = a'd' - b'c'$, $a + d = a' + d'$, then writing

$$a = a - a', \quad b = d + d', \quad f = a' + d, \quad g = a + d', \quad c' = c, \quad h = b,$$

values which satisfy $af + bg + ch = 0$, and taking ξ, η, ζ, ω arbitrary, we have for the matrix Q, which is such that $qQ - Qq' = 0$, the expression

$$Q = \begin{vmatrix} -b\xi - f\eta + h\omega, & h\eta + f\zeta + a\omega \\ -g\eta + a\xi + g\omega, & -c\eta - h\zeta + b\omega \end{vmatrix},$$

depending really on one arbitrary parameter, viz. we may without loss of generality put any two of the coefficients ξ, η, ζ, ω equal to 0.

840.

ON MASCHERONI'S GEOMETRY OF THE COMPASS.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 179–181.]

I HAVE not seen the original of Mascheroni's work, *La Geometria del Compasso*, 8vo, Pavia (1797), but only the French translation, *Géométrie du Compas, par L. Mascheroni, traduite de l'Italien par A. M. Carette*, 2e ed. 8vo Paris, 1828 (Author's Preface, pp. 5–24, and pp. 25–328, with 14 plates containing 108 figures). The title expresses the notion of the work: a straight line is given by means of two points thereof, but is not allowed to be actually drawn; and the problem is, with the compass alone to perform all the constructions of Geometry. Observe that, for the purpose of demonstration, any lines may be imagined to be drawn, and such lines are in fact shown as dotted lines in the figures; but this does not in any wise interfere with the fundamental postulate that the constructions are to be performed with the compass only.

Assuming, then, that a line is in every case given by means of two points thereof, the leading questions are those considered Book VII. "on the intersections of straight lines with circular arcs and with each other," viz. they are (1) to find the intersections of a given line and circle; (2) to find the intersection of two given lines. But in the first problem it is necessary to consider two cases; (A) the general case, where the line does not pass through the centre of the circle, and (B) the particular, but actually more difficult, case, where the line passes through the centre of the circle. It is assumed that the centre of the circle is given: if it is not given, it can be found as afterwards mentioned. It will be convenient to establish the definition of counter-points in regard to a given line: two points, which are such that the line joining them is bisected at right angles by the given line, are said to be counter-points in regard to that line.

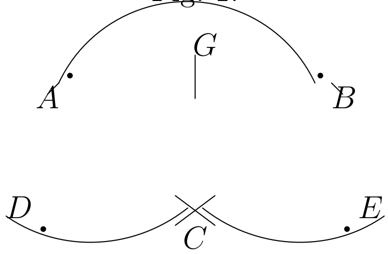
(1), A. Consider a line given by means of two points P, Q; and a given circle, centre C. With centre P and radius PC describe a circle, and with centre Q and radius QC a circle; these meet in C and in a second point D which will be the counter-point of C in regard to the line PQ; hence with centre D and radius = that of given circle, describing a circle, this meets the given circle in two points which are the intersections of the given circle by the line PQ.

The construction fails if the line PQ passes through the centre of the given circle, for then the two auxiliary circles touch at C, or we have D coincident with C. We have therefore considered this case; Q may be taken to be the centre C of the circle; so that we have:

(1), B. The problem is, given a point P, and a circle, centre C: to find the intersections of the circle by the line CP. With centre P, and an arbitrary radius, describe a circle meeting the given circle in points A, B (which are of course counter points in regard to line CP); the circle is divided into two arcs AB and $360 - AB$, and the problem is to bisect each of these two arcs; for the points of bisection are obviously the required intersections of CP with the circle. Hence we have:

(1), C. To bisect a given arc AB of a circle, centre C (fig. 1). With centre A and radius AC describe a circle; and with centre B, and equal radius BC describe a circle; and on these circles respectively, find the points D, E such that $CD = CE = AB$. With centre E and radius EA, describe a circle; and with centre D, and equal radius DB, describe a circle; and let these two circles meet in F; then with centre E (or D) and radius CF describe a circle: this will meet the given circle in the required middle point G of the arc AB.

Fig. 1.



The proof depends on the theorem that in a rhombus the sum of the squares of the diagonals is = sum of the squares of the four sides. We have a rhombus $ACEB$, and therefore

$$(AE)^2 + (BC)^2 = 2(AB)^2 + 2(AC)^2,$$

that is,

$$(AE)^2 = 2(AB)^2 + \text{rad.}^2.$$

But by construction, $EF^2 = (AE)^2$; therefore

$$EF^2 = 2(AB)^2 + \text{rad.}^2, \quad = 2(CE)^2 + \text{rad.}^2.$$

Hence in the right-angled triangle FCE, we have

$$(CF)^2 = (EF)^2 - CE^2, \quad = (CE)^2 + \text{rad.}^2;$$

and by construction, $(EG)^2 = (CF)^2$; that is,

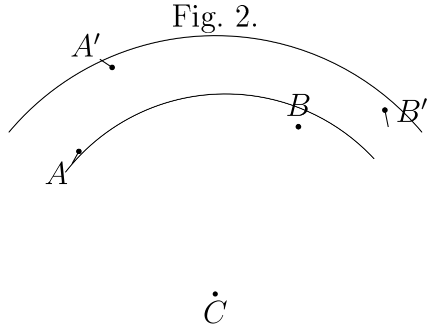
$$(EG)^2 = (CE)^2 + (CG)^2,$$

viz. GEC is a right-angled triangle; that is, CG is at right angles to BCE, or the line CGF bisects the arc AB in G.

For the second problem, (2), to find the intersection of two given lines, we require the solution of the problem to find a fourth proportional to three given distances, and this immediately depends on the following problem.

Problem. In two concentric circles to place chords subtending equal angles. If AB (fig. 2) be a chord of the one circle, then with centres A and B respectively, and an arbitrary radius $AA' = BB'$, describing circles to cut the larger circle in the points A' and B' (A' is either of the two intersections and B' is the intersection lying in regard to CB as A' lies in regard to CA) then clearly $\angle ACB = \angle A'CB'$.

Fig. 2.

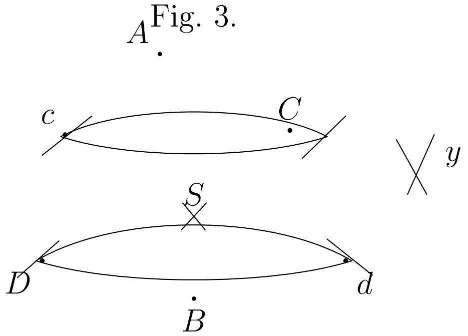


And hence also we have the following problem.

Problem. To find a fourth proportional to three given distances. We have only to take as the given distances the two radii CA, CA' and the chord AB; and then from the similar triangles ACB, $A'CB'$, we have $CA : CA' :: AB : A'B'$, viz. $A'B'$ is a fourth proportional to CA, CA', AB.

We have now the solution of (2) To construct the intersection of two given lines AB and CD (fig. 3). Find c the counter-point of C, and d the counter-point of D, in regard to the line AB

Fig. 3.



and find y such that the distances Cy, dy are $= Dd$, DC respectively; y is in a line with cC, and we have $CydD$ a parallelogram. Find cS a fourth proportional to cy, cd, cC, and with centres c, C respectively and radii each $= cS$ describe circles cutting in the point S; this will be the required intersection of the two lines. In fact, the required point S will be the intersection of the two lines CD, cd: supposing these lines each of them drawn, and also the lines cCy and dy, we have DC parallel to dy, that is, the triangles cdy, cSC are similar to each other or

$$cy : cd :: cC : cS :$$

viz. the distance cS having been found by this proportion, and the point S found as the intersection of the two circles, centres c and C respectively, the point S so determined is the required point of intersection of the given lines AB and CD.

If a circle be given without its centre being known, then taking any three points A, B, C on the circle, and a pair of counter-points D, E of the line AB, and also a pair of counter-points F, G of the line AC, we have obviously the centre of the given circle as the intersection of the lines DE and FG; and the centre can thus be found with the compass only.

It is proper to remark that the problems considered in the present paper are those connecting the theory with ordinary geometry, not the problems which are most readily and elegantly solved with the compass only: a large collection of these are contained in the work, and in particular the twelfth book contains some interesting approximate solutions of the problems of the quadrature of the circle, the duplication of the cube, and other problems not solvable by ordinary geometry.

Cambridge, *March* 19, 1885.

841.

ON A DIFFERENTIAL OPERATOR.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 190, 191.]

WRITE $X = 1+bx+cx^2+\ldots, = (1-\alpha x)(1-\beta x)(1-\gamma x)\ldots$; then by Capt. MacMahon's theorem, any non-unitary function of the roots α, β, $\gamma, \ldots$ is reduced to zero by the operation

$$\Delta, \quad = d_b + bd_c + cd_d + \ldots;$$

for instance, if $(2), = \Sigma\alpha^2 = b^2 - 2c$, we have

$$\Delta(b^2 - 2c) = 2b + b(-2), = 0.$$

We have

$$\Delta X = x^2 + bx^3 + cx^4 + \ldots = xX;$$

and writing, moreover, $X', = b + 2cx + 3dx^2 + \&c.$, for the derived function of X, then

$$\Delta X = x^2 + bx^3 + cx^4 + \ldots = (xX)' \cdot /X$$

We can hence shew that $\Delta\left(\dfrac{X'}{X} - b\right) = 0$; the value is, in fact,

$$\frac{\Delta X' \cdot X - X'\Delta X}{X^2} - \Delta b, \text{ that is, } \frac{(xX)'X - xX \cdot X'}{X^2} - 1,$$

which is

$$= \frac{X + xX' - xX'}{X} - 1, = 0.$$

This is right, for $\dfrac{X'}{X} - b$ is a sum of non-unitary symmetric functions of the roots; in fact,

$$\frac{X'}{X} = \Sigma\frac{-\alpha}{1-\alpha x} = -(1) - (2)x - (3)x^2 - \&c.,$$

or since $b = (1)$, this is $\dfrac{X'}{X} - b = -(2)x - (3)x^2 - \&c.$, a sum of non-unitary functions of the roots.

842.

ON THE VALUE OF $\tan(\sin\theta) - \sin(\tan\theta)$.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 191, 192.]

THE following equation is given p. 59 of the *Lady's and Gentleman's Diary* for 1854:

$$\tan(\sin\theta) - \sin(\tan\theta) = \tfrac{1}{30}\theta^7 + \tfrac{29}{756}\theta^9 + \&c.$$

Write in general

$$\begin{aligned} X &= \theta + A\theta^3 + B\theta^5 + C\theta^7 + D\theta^9 + \ldots, \\ Y &= \theta + A'\theta^3 + B'\theta^5 + C'\theta^7 + D'\theta^9 + \ldots. \end{aligned}$$

Then, as far as θ^9, we have

$$\begin{aligned} X + A'X^3 + B'X^5 + C'X^7 + D'X^9 = {} & \theta + A\theta^3 + B\theta^5 + C\theta^7 + D\theta^9 \\ & + A'(\theta^3 + 3A\theta^5 + 3(A^2+B)\theta^7 + (A^3 + 6AB + 3C)\theta^9) \\ & + B'(\theta^5 + 5A\theta^7 + (10A^2 + 5B)\theta^9) \\ & + C'(\theta^7 + 7A\theta^9) \\ & + D'(\theta^9) \\ = {} & \theta \\ & + \theta^3(A + A') \\ & + \theta^5(B + 3AA' + B') \\ & + \theta^7(C + 3A^2A' + 3BA' + 5AB' + C') \\ & + \theta^9(D + A'(A^3 + 6AB + 3C) + 5B'(2A^2 + B) + 7AC' + D'), \end{aligned}$$

and hence

$$\begin{aligned} &X + A'X^3 + B'X^5 + C'X^7 + D'X^9 \\ &-Y - A'Y^3 - B'Y^5 - C'Y^7 - D'Y^9 \\ &= \theta^7\{3AA'(A - A') + 2(AB' - A'B)\} \\ &+\theta^9\{AA'(A^2 - A'^2) + 6AA'(B - B') + 4(AC' - A'C) + 10(A^2B - A'^2B')\}. \end{aligned}$$

Now let

$$\begin{aligned} &X = \sin\theta = \tfrac{1}{2}\theta - \tfrac{1}{6}\theta^3 + \tfrac{1}{120}\theta^5 - \tfrac{1}{5040}\theta^7 + \ldots; \\ &A = -\tfrac{1}{6}, \quad B = \tfrac{1}{120}, \quad C = -\tfrac{1}{5040}, \\ &Y = \tan\theta = \tfrac{1}{2}\theta + \tfrac{1}{3}\theta^3 + \tfrac{2}{15}\theta^5 + \tfrac{17}{315}\theta^7 + \ldots; \\ &A' = \tfrac{1}{3}, \quad B' = \tfrac{2}{15}, \quad C' = \tfrac{17}{315}: \end{aligned}$$

we have therefore

$$\begin{gathered} AA' = -\tfrac{1}{18}, \quad A - A' = -\tfrac{1}{2}, \quad A + A' = \tfrac{1}{6}, \quad B - B' = -\tfrac{1}{8}, \\ AB' - A'B = -\tfrac{1}{45}, \quad AC' - A'C = -\tfrac{17}{1890}, \quad A^2B - A'^2B' = \tfrac{19}{2520}. \end{gathered}$$

Hence

$$\begin{aligned} &\text{coeff. } \theta^7 = \tfrac{1}{12} - \tfrac{2}{45}, \quad = \tfrac{1}{30}, \\ &\text{coeff. } \theta^9 = \tfrac{1}{216} + \tfrac{1}{24} - \tfrac{17}{945} + \tfrac{19}{504}, \quad = \tfrac{29}{756}; \end{aligned}$$

and the required equation is thus verified.

843.

ON THE QUADRIQUADRIC CURVE IN CONNEXION WITH THE THEORY OF ELLIPTIC FUNCTIONS.

[From the *Mathematische Annalen*, t. XXV. (1885), pp. 152–156.]

I CALL to mind that, if we have on a line two points $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$, and through the line two planes

$$Ax + By + Cz + Dw = 0, \quad A'x + B'y + C'z + D'w = 0,$$

then

$$\begin{aligned} B\gamma' - B'\gamma : \gamma\alpha' - \gamma'\alpha : \alpha\beta' - \alpha'\beta : \alpha\delta' - \alpha'\delta : \beta\delta' - \beta'\delta : \gamma\delta' - \gamma'\delta \\ = AD : BD : CD : BC : CA : AB, \end{aligned}$$

and that, putting each of these two equal sets of values

$$= a : b : c : f : g : h, \quad \text{or} \quad = A : B : C : D : BC : CA : AB,$$

then the quantities a, b, c, f, g, h which it is easy to see satisfy the relation $af + bg + ch = 0$, are called the "six coordinates" on the line; we call them the six coordinates of the line, as also the coordinates of either of the six line-axiotomical equations which is a single relation between these axes, the so-called axiotomical contains a single relation between them. Or, in the language of the modern geometry, the determination of the particular line. See my paper "On the six coordinates of a line," *Camb. Phil. Trans.* t. XI. (1869), pp. 290–323, [435].

I consider for a moment the quadric surface

$$\alpha x^2 + \beta y^2 + \gamma z^2 + \delta w^2 = 0,$$

and I proceed to shew that, if (α, b, c, f, g, h) are the coordinates of a generating line on the surface, then either $a = \sqrt{f}$, $b = \sqrt{g}$, $c = \sqrt{h}$, or else $a = -\sqrt{f}$, $b = -\sqrt{g}$, $c = -\sqrt{h}$; the one or other system of equations according as the line belongs to the one or other system of generating lines.

We satisfy the equation by

$$\alpha x + by + \theta z + \theta w = 0, \quad x = \frac{1}{\phi}y - \frac{1}{\phi}z + \theta w = 0,$$

where θ is an arbitrary parameter: hence these equations determine a generating line of the surface: and the coordinates of this line are

$$a, \quad b, \quad c, \quad f, \quad g, \quad h$$

$$= -1\left(\theta - \frac{1}{\phi}\right), \quad -\left(\theta + \frac{1}{\phi}\right), \quad \theta\phi, \quad -\left(\theta - \frac{1}{\phi}\right), \quad \theta + \frac{1}{\phi}, \quad -\phi;$$

viz. these values give $a = \sqrt{f}$, $b = \sqrt{g}$, $c = \sqrt{h}$.

Similarly we satisfy the equation by

$$\alpha x + by + \phi z - \phi w = 0,$$

$$x = \frac{1}{\phi}y - \frac{1}{\phi}z - \phi w = 0,$$

where ϕ is an arbitrary parameter: hence these equations also determine a generating line of the surface: and the coordinates of this line are

$$a, \quad b, \quad c, \quad f, \quad g, \quad h$$

$$= -1\left(\phi - \frac{1}{\phi}\right), \quad \phi + \frac{1}{\phi}, \quad -\phi^2, \quad -\left(\phi - \frac{1}{\phi}\right), \quad \phi + \frac{1}{\phi}, \quad -\phi^2;$$

viz. these values give $a = -\sqrt{f}$, $b = -\sqrt{g}$, $c = -\sqrt{h}$.

If for x, y, z, w we write $x\sqrt{f}$, $y\sqrt{g}$, $z\sqrt{h}$, $w\sqrt{f}$ respectively, then we have the theorem that, for the quadric surface

$$pa^2 + qy^2 + rz^2 + sw^2 = 0,$$

the coordinates (a, b, c, f, g, h) of a generating line are such that

$$\frac{a}{\sqrt{f}} = \pm\sqrt{\frac{pa}{qb}}, \quad \frac{b}{\sqrt{g}} = \pm\sqrt{\frac{q}{r}}, \quad \frac{c}{\sqrt{h}} = \pm\sqrt{\frac{r}{s}};$$

the signs being all + or all −, according as the line belongs to the one or other of the systems of generating lines.

Take $(\alpha', b', c', f', g', h')$ for the coordinates of an arbitrary line, and write

$$p, q, r, s = q\beta(f'b' + b'h'), q\beta(g'a' + a'g'), r\beta(g'a' + a'g'), s\beta(c'a' + a'c'); = a'fbcgh',$$

$$a'g'bsr - b'sf'gh' + c'gh'a' + cha'b'g'd = 0,$$

which is a section having the line (a', b', c', f', g', h') for a generating line. To verify this, observe that for the line in question we have

$$\begin{aligned} Ky - qx + zw &= 0, \\ -Kx + qy + zw &= 0, \\ yg + bg' \cdot x + xw &= 0, \\ -x'g + b'p \cdot y + z'w &= 0, \end{aligned}$$

equivalent of course to two independent equations: these give $K(x = b'y \cdot k)w$, $Ky = pa \cdot kw$ and so on, where values substituted in the quadric equation give it identically.

And for the last-mentioned quadric surface that, for the system $a = \sqrt{f}$, $b = \sqrt{g}$, $c = \sqrt{h}$, of the conditions for a generating line, the equation reduces itself to

$$af + bg + ch = 0,$$

where the signs $=$ correspond to the system a, b, c, g, h, f, and the sign $-$ to the other system.

Taking R as origin, the equations of the conic are

$$Kx = py - qz + zw = 0,$$

$$-Kx + qy - rz + sw = 0,$$

equivalent of course to two independent equations of the form

$$\alpha\beta x + \alpha'\beta' y + \beta'\alpha c + s\alpha\beta' = 0, \quad \text{equation of a generating line,}$$

which is of course not to be confounded with the equation of a generating line.

Considers now the quadriquadric curve

$$(x, y, z, w)(a', b', c', f', g', h') = 0, \quad (x, y, z, w)(\alpha', \beta', \gamma', \delta') = 0,$$

where the quadric forms a'', a'', are connected with each other by the relation $a'\beta'a' + a''b'a' + a''c'a' + a''d'a' = 0$.

If, then, for an arbitrary line (a', b', c', f', g', h'), we have $aa' + bb' + cc' + ff' + gg' + hh' = 0$, this relation is also one of the conditions for the relation between any two lines: any one whatever of the four lines through any point of the curve corresponds to the two systems respectively, of a properly determined quadric surface: it is a property of this curve that, of any four lines through any one of the points whatever of the system, those one is symmetric (whatever be the value), in the other one with the system, and the other two are unconnected.

Writing in the equations $a = 1$, we have in particular the quadriquadric curve

$$y^2 + b'\beta' z + \beta = 0, \quad x^2 + b'' y + c'' z + d'' w = 0,$$

where the second is the symmetric definition of the system through the equation $g + h = 0$. Consider on the curve four points belonging to the arguments u_1, u_2, u_3, u_4 respectively;

41 – 2

and write for shortness x_1, y_1, z_1, w_1 for the sn, cn, and similarly for x_2, y_2, z_2, w_2. It appears by Abel's theorem that the condition in order that the four points may be in a plane is $u_1 + u_2 + u_3 + u_4 = 0$; viz. when this equation is satisfied we have

$$\begin{vmatrix} x_1, & y_1, & z_1, & w_1 \\ x_2, & y_2, & z_2, & w_2 \\ x_3, & y_3, & z_3, & w_3 \\ x_4, & y_4, & z_4, & w_4 \end{vmatrix} = 0,$$

or writing

$$\begin{aligned} & a, \quad b, \quad c, \quad f, \quad g, \quad h \\ &= x_4y_2 - x_2y_4, \quad x_3y_1 - x_1y_3, \quad z_4w_2 - z_2w_4, \quad z_3w_1 - z_1w_3, \\ & \quad a_2, \quad b'', \quad c'', \quad f', \quad g', \quad h', \\ &= x_4y_4, \quad x_3y_3, \quad z_4w_4, \quad z_3w_3, \quad y_3z_3, \quad x_4w_4, \end{aligned}$$

this equation is

$$af + bg + ch = 0;$$

or by what precedes, it appears that not only is this so, but that we have separately

$$af^2 + cf'' = 0, \quad bg' + b'g = 0, \quad ch' + ckh = 0,$$

viz. it follows from Abel's theorem that, when the arguments u_1, u_2, u_3, u_4 are connected by the equation $u_1 + u_2 + u_3 + u_4 = 0$, then each of these three equations holds good.

I assume in particular $u_1 = 0$, $u_2 = -u$, so that $x_2 = x$, $u_2 = -u$, $y_2 = y$, $z_2 = z$, $w_2 = w$, $x_3 = 0$, $y_3 = b_2$, $w_3 = a_2$, $w_2 = c_2$; we have $x_4 = a$, $b_4 = b_4 = b_2 = c_2$, $w_4 = d_4 = c_2$, $w_4 = c_4$,

$$\begin{aligned} & \alpha, \quad b, \quad c, \quad f, \quad g, \quad h \\ &= a_2 - d_2x, \quad -x_1 = -a_2x = -a_2x_2 = -x_2x_2x_1, \quad x = b_2y, \quad -x_2y = d_3y, \quad d_2x_3y = -x = b_4y - 1, \end{aligned}$$

and the four equations become

$$\begin{aligned} \frac{a - d_2x \cdot d_4z + b_2}{x} \cdot d_2w &= 0, \\ -x = -d_2y + a_2x &= 0, \\ \frac{d_2 - z_2 \cdot b_2y - 1}{c} + \frac{d_2x - z_2 \cdot a_2y - 1}{c_2} &= 0, \end{aligned}$$

equations which may be written

$$\frac{a - d_2x}{a_2x} = \frac{w_2 \cdot d_4z + d_2y}{a_2x} \cdot \frac{-x}{-z} \cdot \frac{c_3x}{c_3} = \frac{a_2x + d_2y}{c_3}, \qquad \frac{x}{a_2x - d_2y} = \frac{c_2}{c_3} \cdot \frac{a_2x - d_2y}{c_3},$$

so that, adding these equations, we must have an identity.

Representing the second and third of the last-mentioned equations by

$$\frac{a+1}{x} = C, \quad \frac{a-1}{x} = D,$$

we have

$$a = 1 - Cx, \quad d = 1 + Dx,$$

from either of which we can obtain a rationally; viz. the first equation gives

$$1 - a = 1 - 2Cx + C^2x^2,$$

that is, $x = \dfrac{2C}{1+C^2}$, whence

$$a = \frac{1-C^2}{1+C^2}, \quad d = \frac{1+C^2+2CD}{1+C^2};$$

and similarly from the second equation $a = -\dfrac{1}{2D}\cdot$ whence also

$$a = \frac{b'z+2P-2CD}{D^2+D^2}, \quad d = \frac{D^2-1}{D^2+D^2}.$$

Substituting for C its value, the first-mentioned value of a becomes

$$a = \frac{(a_2-z_4)(d_2x-d_2z)}{1-2x_2x_4z-z_2z_4z_2x_4},$$

which is a form due to Cayley.

For the like values of b and d we easily verify; we in fact have

$$x_3 = sn(u_3+u_4) = \frac{u_2u_3\cdot d_2u_3 + a_3u_2\cdot d_3u_2}{1-k^2u_2u_3\cdot a_3u_2\cdot a_3\cdot u_3},$$

$$= \frac{a_2u_3\cdot d_3u_3 + x_2u_3\cdot a_2u_3}{a_3u_2\cdot a_2 + k^2x_2x_3y_2y_3},$$

we say *Elliptic Functions*, p. 63; or calling these values

$$= \frac{N_1}{D_1} = \frac{N_2}{D_2} = \frac{N_3}{D_3} = \frac{N_4}{D_4},$$

the above value is

$$x = \frac{N_1-N_2}{D_1-D_2},$$

which is right.

Cambridge, *28 June*, 1884.

844.

ON A THEOREM RELATING TO SEMINVARIANTS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XX. (1885), pp. 212, 213.]

I FIND among my papers the following example of a general theorem relating to seminvariants. Write:

$$\begin{aligned}\Theta &= 4b_8 + a_2 b_6 + a_2^2 b_4 + a_2^3 b_2 + 4a_2^4 b_0 + 7a_4 b_4 + 16a_4^2 b_0,\\ \Omega &= \quad b_8 + 3a_2 b_6 + 5a_2^2 b_4 + 4a_4 b_4 + 5a_2^2 b_2 + 9a_4 a_2 b_2 + 7a_4^2 b_2;\end{aligned}$$

then from a seminvariant S of the degree δ and weight w, by means of the operation Θ we obtain a seminvariant S of the degree $\delta + 1$ and weight $w + 8$; and by operating with Ω we obtain a combination of these by operating on S with the resulting combination $2w\Omega - 8\Omega$, which combination is of the degree $w + 1$, but only of the degree $w + 8$; via. operating with $2w\Omega - 8\Omega$, we obtain a new seminvariant of the same degree but of a weight increased by unity.

The example relates to the under-mentioned seminvariant S of the degree 5 and weight 7. Write

b_0	$+1$
b_2	-7
b_4	-9
b_6	$+9$
b_8'	-123
b_8	-10
bf'	$+39$

ΘS	ΩS	$\Theta\Theta S$	$\Theta\Omega S$	$\Omega\Theta S$	$\Omega\Omega S$						

[*The two-page paper concludes with a numerical verification table demonstrating the action of* Θ *and* Ω *on the chosen seminvariant S; for the table of differences and the proof that* $\Theta\Omega - \Omega\Theta = 0$ *on S, see the original Quarterly Journal, vol. XX, p. 213.*]

The columns above ΘS (where observe that, to operate with the b_8, of Θ, we restore the proper power of a_2, reading S as being $= 1.b^4 + 7a^3b_2$, and proceeding ultimately $\Theta S = \frac{1}{8}\Omega S = -55\Omega$; and the sum of these, which is a seminvariant, degree 5, weight 7, also $125''$ and 145); and the sum of these, which is a seminvariant, degree 5, weight 7, also $125''$ and 145; the upper portion comprises proof of the theorem $S = 1.b_4 + 7a^3b_2$; instead of giving this sum, I have added a column equal to $\frac{1}{16}(\Theta\Omega - \Omega\Theta)S$ this gives also the sum of the above seminvariants, viz. when this equation is the foregoing direct process from the given seminvariant S of the degree 5 and weight 7.

845.

ON THE ORTHOMORPHOSIS OF THE CIRCLE INTO THE PARABOLA.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XX. (1885), pp. 213–220.]

IT is remarked by Schwarz (see his Memoir "Ueber einige Abbildungsaufgaben," *Crelle*, t. LXX. (1869), pp. 105–120, p. 113), that the circle $x^2+y^2-1=0$ can be orthomorphosed into the parabola $y^2+4(1-x)=0$ by the equation

$$x'(x'+iy') = \log x + iy'(x+iy);$$

viz. this equation establishes a $(2,1)$ relation between the points interior to the circle and those interior to the parabola, which, qua relation between $x'+iy'$ and $x+iy$, will be orthomorphic, that is, infinitesimal elements of the one area will correspond to similar infinitesimal elements of the other area. The diameter $y'=0$ of the circle is transformed into the axis $y=0$ of the parabola, and figs. 1, 2 are symmetrical on the two sides of these lines respectively; we may therefore consider only the transformation

Fig. 1.

[Figure: upper semicircle in the $x'y'$-plane bounded by the diameter on the x'-axis]

Fig. 2.

[Figure: upper half of the parabola in the xy-plane bounded by the axis $y=0$]

of the upper semicircle into the upper half of the parabola; we have (see figs. 1 and 2) A' corresponding to the vertex A of the parabola, B' to the point at infinity on the infinite branch AB of the parabola, the semicircular boundary $B'C'A'$ to the boundary BCA of the parabola, the highest point (and any point C' between the vertex A and B' on the parabola, the highest point C corresponding to the point C' between the vertex and the semi-latus rectum.

Fig. 3.

[Figure: semicircle with concentric circles and radii, and confocal parabolas with axial system through focus, illustrating the orthomorphic correspondence]

We may divide the circle by concentric circles and by radii; the corresponding curves for the parabola will be ovals and radials from the focus, the curves of each system being transcendental curves. Or we may divide the parabola by means of two systems of confocal parabolas; the corresponding curves for the circle will be (see fig. 3) two systems of orthotomic licences, those of the one system having B' for a crossnode, and those of the other system having B' for an asymode.

Writing log tan to denote the hyperbolic logarithm of the tangent, that if ϕ, ϕ' are such that

$$\phi = \log\tan(\tfrac{1}{4}\pi + \tfrac{1}{2}\phi'),$$

this equation, as is known, may also be presented in the form

$$i\phi' = \log\tan(\tfrac{1}{4}\pi + \tfrac{1}{2}i\phi),$$

$$i\tan\tfrac{1}{2}\phi' = \tan\tfrac{1}{2}i\phi,$$

or, what is the same thing,

$$\tan\tfrac{1}{2}\phi' = \tanh\tfrac{1}{2}\phi,$$

where observe that, as ϕ' increases from 0 to $\frac{1}{2}\pi$, ϕ increases from 0 to ∞, and that always $\phi > \phi'$.

We can now establish as follows the three correspondences $A'O$ with $A'C$, $C'B$ with OB, and $A'CB$ with $A'C'B'$:

(1) $x'(x'+iy') = \frac{2\phi}{i}$, $\quad x'+iy' = \tan\frac{1}{2}\phi'$;

that is,

$$x = -\frac{2\phi'}{\pi}, \quad y = 0; \quad x' = \tan\tfrac{1}{2}\phi', \quad y' = 0;$$

ϕ is given as A, A'; ϕ' is given as $\theta = 0$, C, C';

(2) $x'(x'+iy') = -\tan^2\frac{1}{2}\phi'$, $\quad x'+iy' = i\tan\frac{1}{2}\phi'$;

that is,

$$x = -\tan^2\tfrac{1}{2}\phi', \quad y = 2\tan\tfrac{1}{2}\phi'; \quad x' = 0, \quad y' = \tan\tfrac{1}{2}\phi';$$

$\phi = 0$ gives A, A'; $\phi = \frac{1}{2}\pi$ gives B, B';

(3) $x'(x'+iy') = \frac{2i\phi}{\pi}$, $\quad x'+iy' = e^{i\phi'}$;

that is,

$$x = 1 - \frac{2\phi}{\pi}, \quad y = \frac{2\phi}{\pi},$$

the parabola; and

$$x' = \cos 2\phi', \quad y' = \sin 2\phi',$$

the circle;

$$\phi = 0 \text{ gives } A, A'; \quad \phi = \tfrac{1}{2}\pi \text{ gives } B, B'.$$

Observe then for points in OA', $A'B'$, equidistant from O, we have $x' = \tan\frac{1}{2}\phi'$, $x' = -\tan\frac{1}{2}\phi'$, and corresponding hereto we have points in OA, OB on the axis of the parabola, the values of x being $x = 1 - \frac{2\phi}{\pi}$, $x' = -1 - \frac{2\phi}{\pi}$, viz. the negative value is always the greater.

Observe further that, to the points $(x', y') = (\cos 2\phi', \sin 2\phi')$ and $(x', y') = (-\cos\frac{1}{2}\phi', 0)$ on the circle, and on the radius OB', correspond the points

$$(x, y) = \left(1 - \frac{2\phi}{\pi}, \frac{2\phi'}{\pi}\right) \text{ and } (x, y) = \left(-\frac{2\phi}{\pi}, 0\right);$$

on the parabola and on the axis OB respectively; the axial distance of these two points is $\left(1 - \frac{2\phi}{\pi} + \frac{2\phi}{\pi}\right) = 1$, the radius of the circle, or the distance OA of the vertex and focus of the parabola; this is a rather curious theorem.

The function $\log\tan(45+\frac{1}{2}\alpha\phi')$ is tabulated, Legendre, *Théorie des Fonctions Elliptiques*, t. II. table II. pp. 256–259; viz. writing as above ϕ for the argument, we have hereby the value of ϕ, we obtain ϕ from 90 to 90 at intervals of $30'$ and to 12 decimals, and with 6th differences. Observe that ϕ' is thus given in degrees and minutes as a circular arc, ϕ as a number; it is convenient to have ϕ and ϕ' each as arcs, or such as numbers, the conversion may of course be made by means of a table of the lengths of circular arcs, and I have calculated the Table which I give at the end of the present paper.

I had previously calculated for the correspondence $A'O$, OB' and $A'CB$ with $A'O$, OB and ACB, the following values, at irregular intervals suited to the construction of a figure:

Circle		Parabola	
$x'OB'$	Ordinate $\div R$	$x'O$	OB
0	0	0	0
30	.5176	.1641	$-.1641$
60	1.0000	.5851	$-.5851$
80	1.2856	1.0470	-1.0470
100	1.5320	1.6262	-1.6262
130	1.8126	2.5817	-2.5817
150	1.9319	3.4131	-3.4131
160	1.9696	3.8967	-3.8967
170	1.9924	5.5942	-5.5942
175	1.9981	4.0024	-4.0024
178	1.9994	4.6231	-4.6231
180	0	$-\infty$	∞

Write in general

$$x'(x'+iy') = \rho + i\sigma, \quad x'+iy' = \tan\tfrac{1}{2}(\rho+i\sigma);$$

viz. considering x, y as given, find these ρ, σ by the equations $x = p\sigma'\pi$, $y = 2\sigma\pi$, and then writing $\frac{1}{2}\rho = \theta$, $\frac{1}{2}\sigma = \phi$, we have

$$x'+iy' = \tan(\theta+i\phi) = \tan(\theta+i\phi) = \frac{P+iQ}{1-iPQ};$$

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where

$$P = \tan\tfrac{1}{2}\rho\theta, \quad = \tan\tfrac{1}{2}\theta;$$

$$Q = \tfrac{1}{i}\tan\tfrac{1}{2}i\phi, \quad = \tanh\tfrac{1}{2}\phi, \quad = \tfrac{1}{i}\tan\tfrac{1}{2}i\phi, \quad = \tanh\tfrac{1}{2}\phi;$$

whence, if we introduce ϕ' connected with ϕ by the equation $\phi = \log\tan(\frac{1}{4}\pi + \frac{1}{2}\phi')$ as before, we have $Q = \tan\frac{1}{2}\phi'$, and the formula is

$$x'(x' + iy') = \frac{P + iQ}{1 - iPQ}, \quad P = \tan\tfrac{1}{2}\theta, \quad Q = \tan\tfrac{1}{2}\phi',$$

giving the values of x', y'.

It is clear that we have

$$x'(x' + iy') = \frac{P - iQ}{1 + iPQ},$$

and thence

$$x'(x' + y'^2) = \frac{P + Q}{1 + P^2Q^2}.$$

Hence, to the circle $x'^2 + y'^2 = c^2$, corresponds in the parabola the curve

$$P + Q = c^2(1 + P^2Q^2),$$

where $p + iq = \tan(\rho + i\sigma)$, $P = \tan\frac{1}{2}\rho$, $Q = \tan\frac{1}{2}\phi'\ \tan\ \frac{1}{2}\sigma$. This is a complicated transcendental equation, and I do not see any convenient way of tracing the curve. The set of curves satisfy the differential equation

$$\frac{PdP + QdQ}{P + Q} = -\frac{PQ(QdP + PdQ)}{1 + P^2Q^2},$$

where dP, dQ are given in terms of $d\rho$, $d\sigma$ by

$$PdP(1 + Q^2) - QdQ(1 + P^2) = 0,$$

that is,

$$\frac{dP}{d\sigma} = \tfrac{1}{2}\sec^2\tfrac{1}{2}\rho, \quad \frac{dQ}{d\sigma} = \tfrac{1}{2}\sec\phi'\sec^2\tfrac{1}{2}\phi'.$$

We have $p = \frac{2\rho}{\pi}$, and thence we have to take, as elements of the several curves, $\frac{1}{2}d\rho + i\frac{1}{2}d\phi = 0$. Combining these equations, we have

$$dP\frac{dQ}{Q} \cdot dQ + (1 + P^2) - q(1 + Q^2),$$

and thence

$$\rho^2 dP(1 + Q^2) - qQdQ(1 + P^2) = 0,$$

as an equation in the locus of the normals in question. If p and q are small, then putting for a moment $\frac{1}{2}p = a$ for shortness, we have

$$P = a\rho + \tfrac{1}{3}a\rho^2, \quad Q = a\sigma + \tfrac{1}{3}a\sigma^3,$$

and the equation becomes

$$p^2(1 + \tfrac{1}{3}a\sigma^2)^2 - q^2(1 - \tfrac{1}{3}a\sigma^2)^2 - q^2(1 + \tfrac{1}{3}a\rho^2)^2 = 0,$$

that is,

$$p^2 - q^2 + (a\rho^2 + \alpha^2)(p^2 + q^2)(p + q) = 0,$$

writing this in the form

$$p^2 - q^2 = -\frac{\pi^2}{16}(p^2 + q^2)\rho^2 + a^2(p^2 + q^2),$$

we find $x = \frac{\pi^2}{8}(x^2 + 2y^2) = 0$, or say $p^2 = \frac{\pi^2 y^2}{8}$ as the form of the meanits in the neighbourhood of the focus. Or, viz. the meanits for all of these, as might have been expected, are the kit-shaped (or negative side) of the curve.

I have been desirous to find a more general law 0 = between the meanits in the neighbourhood of the focus. We may say $\rho = -\frac{2\pi}{\pi}$, as in the form of the meanits in the neighbourhood of the focus.

Corresponding hereto in the circle, writing for a moment

$$x'(x' + iy') = \rho i,$$

we have

$$\tan \tfrac{1}{2}\rho i = \frac{P + iQ}{1 - iPQ},$$

whence

$$\rho i = \rho - i \cdot (1 + p^2/\pi^2),$$
$$\rho = P^2(1 - Q^2)(1 + pP^2);$$

whence eliminating P^2 and Q successively, we may say

$$\rho^2 + \rho^2 = a^2\left(P^2 - \frac{1}{P^2}\right) - 1 = 0,$$

$$\rho^2 + \rho^2 = a^2\left(\frac{1}{Q^2} + 1\right) - 1 = 0,$$

say, for shortness, then $\rho^2 + 2\rho\rho\rho' + a' = 0$, and $\rho^2 + \rho\rho - 2\rho\rho' = 0$. Introducing the polar coordinates x', y' then

$$x' = r\cos\theta', \quad y' = r\sin\theta',$$

the equations then become

$$r^2 = 2\alpha\rho'\sin\tfrac{1}{2}\theta' + 1,$$
$$\rho^2 = -2\alpha\rho'\cos\tfrac{1}{2}\theta' + 1,$$

the polar coordinates r, θ'.

Consider the quadricuspidal curve

$$r' = Ax^2 + Bx^3 + Cx + Dx^2 + Ex,$$

where $A = 4(1 + 2P^2)(1 + p^2)$, $B = \frac{1}{2}p^2$, $C = 4 + ix^2(1 - 2P^2)$, $D = +\frac{1}{2}p^2(1 + 4y)$, the coordinates of two lines meeting each side, and consequently it entirely contains the curve $\rho i, \rho i, \rho i$. They agree, as they should do, with a foregoing result,

$$\rho^2 = 0,$$

$$\rho^2 + Dy^2(t' + \tfrac{1}{2}h\theta'\rho') - 6 \cdot \tfrac{1}{2}\rho^2 + \tfrac{1}{2}\rho^2 + \tfrac{1}{2}\rho = 0,$$

and it would of course be possible to express the remaining forms 213, 642, 662, 441, 432, and 333 in terms of $B = 810$ and $D = 630$. But observe that the speciality of which belong to two lineapas having each of them B' for a node and O for a focus, and which of course cut each other at right angles; see fig. 4, which is a more

Fig. 4.

[Figure: two orthogonal families of lemniscates with node B' and focus O, with a generic point indicated]

diagram. In fact, omitting for convenience the accents, but recollecting always that the curves belong to the figure of the circle, the first equation gives

$$(x-1)^2 + 4ax^2 \cdot \cos^2 \tfrac{1}{2}\theta', \quad = 2a\rho'(1+\cos\theta) = 2a\rho'(x+r),$$

that is,

$$(r^2 + 1 - 2x)x^2 = 4a\rho'(x+r)^2.$$

Transferring to the origin B', we must write $x+1$ for x, and then

$$r'^2 = (x+1)^2 + y^2.$$

The equation thus becomes

$$[x^2 + y^2 - 2(\alpha x + 1) + 3(\alpha x + 1)]^2 = 4a^2(x+1)(x+y^2+2x+1)),$$

that is,

$$[x^2 + y^2 - 2(\alpha x + 1)]^2 = 4a^2(x^2+y^2)(x^2+y^2),$$

or say

$$(x^2+y^2)^2 + 4(\alpha x + 1)(x^2+y^2) + 4(x+1)(x^2+y^2) + 4(\alpha x + 1)\alpha^2 y^2) = 0,$$

showing that the point B' is a crunode, the tangents there being $x \pm \alpha y$.

Writing in the equation $y = 0$, we have $x^2 = 0$, the crunode, and

$$x = 2\sqrt{(\alpha'+1)}\sqrt{(\alpha'+1)} \pm 1.$$

the product of these two values, $\alpha = 4(\alpha+1)$, that is, $4 = \left(P + \frac{1}{p}\right)^2$, via. one value is greater, the other less, than 2. We see also that

$$2\sqrt{(\alpha+1)}\sqrt{(\alpha'+1)} - 1$$

the smaller root, is greater than 1. The curve corresponding to a parabola $y^2 = 4p(c-x)$ is thus a crunodal lineapas, the crunode at B', and the loop lying wholly within the circle. Moreover, the loop includes within the centre O of the circle.

The other curve is in like manner

$$[r^2 + 1 - 2x]^2 = 4(1-x)^2,$$

viz. transforming to the origin B', and therefore writing $x = 1$ for x, and similarly for x_n, and x_n. In support by Abel's theorem that the condition in order that the four points may be in a plane is $x = x_n + x_n + 1 = 0$; viz. when this equation is satisfied we have

$$(r^2 + 1 - 2x)x^2 = 4ax^2(x^2+1)^2,$$

that is,

$$[x^2 + y^2 - 2x(x+1)]^2 = 4ax^2(x^2+y^2),$$

or

$$(x^2+y^2)^2 - 4(\alpha+1)(x^2+y^2)x + 4(\alpha+1)x^2 + 4\alpha^2(x^2+2y^2-\alpha^2y^2) = 0,$$

showing that the point B' is a crunode, the tangents being $x \pm \alpha y$.

Writing in the equation $y = 0$, we have $x^2 = 0$, the crunode, and

$$x = 2\sqrt{(\alpha+1)} \cdot x \pm 1; \quad 4\alpha = -1, \quad x = -1,$$

the product of these values is 4α, that is, $4\alpha = -\left(P - \frac{1}{p}\right)^2$, via. one value is greater, the other less, than 2. We see also that

$$2\sqrt{(\alpha+1)}\sqrt{(\alpha+1)} - 1$$

is positive and less than 1; the curve is thus an acnodal lineapas, having B' for the acnode and cutting the axis outside the circle at the points $x = \alpha + 2\sqrt{(\alpha+1)}$. Conversely, considering the former curve as given by the expressions of the coordinates in terms of a parameter ϕ, the determination of the like expressions for the latter curve would appear to be a problem of very great difficulty.

The Table above referred to.

[Table giving the correspondence between $\phi = \log\tan(45 + \frac{1}{2}\phi')$ for ϕ' from 0 to 90 at intervals of 1. The full numerical table from the original paper, with values to seven decimal places, is reproduced below.]

ϕ'	$\phi = \log\tan(45 + \frac{1}{2}\phi')$	ϕ'	ϕ	ϕ'	ϕ
0	0	46	.9028545	50	.9962755
1	.0174553	47	.9249187	51	1.0381004
2	.0349066	48	.9457040	52	1.0584952
3	.0523599	49	.9657500	53	1.0792677
4	.0698132	50	.9852280	54	1.1005044
5	.0872665	51	1.0036321	55	1.1222263
6	.1047198	52	1.0216162	56	1.1444466
7	.1221730	53	1.0392060	57	1.1672144
8	.1396263	54	1.0566175	58	1.1905610
9	.1570796	55	1.0738249	59	1.2145202
10	.1745329	56	1.0908669	60	1.2391411
11	.1919862	57	1.1077612	61	1.2644777
12	.2094395	58	1.1244912	62	1.2905931
13	.2268928	59	1.1410615	63	1.3175552
14	.2443461	60	1.1573912	64	1.3454387
15	.2617994	61	1.1733035	65	1.3743253
16	.2792527	62	1.1887844	66	1.4042947
17	.2967060	63	1.2036336	67	1.4354352
18	.3141593	64	1.2178358	68	1.4678434
19	.3316126	65	1.2312718	69	1.5016262
20	.3490659	66	1.2438139	70	1.5369024
21	.3665192	67	1.2555401	71	1.5738033
22	.3839724	68	1.2663275	72	1.6124704
23	.4014257	69	1.2761519	73	1.6530834
24	.4188790	70	1.2848920	74	1.6958685
25	.4363323	71	1.2924327	75	1.7410341
26	.4537856	72	1.2986277	76	1.7887898
27	.4712389	73	1.3033236	77	1.8394554
28	.4886921	74	1.3064565	78	1.8933630
29	.5061454	75	1.3078560	79	1.9508587
30	.5235987	76	1.3074457	80	2.0123399
31	.5410520	77	1.3050447	81	2.0782854
32	.5585053	78	1.3004563	82	2.1492283
33	.5759585	79	1.2934613	83	2.2258028
34	.5934118	80	1.2838174	84	2.3087094
35	.6108651	81	1.2712504	85	2.3989343
36	.6283184	82	1.2554543	86	2.4977049
37	.6457717	83	1.2360880	87	2.6066908
38	.6632250	84	1.2127740	88	2.7282169
39	.6806782	85	1.1850968	89	2.8657689
40	.6981315	86	1.1525987	90	∞
41	.7155848	87	1.1147993		
42	.7330381	88	1.0712310		
43	.7504914	89	1.0214444		
44	.7679447	90	∞		
45	.7853982				

846.

A VERIFICATION IN REGARD TO THE LINEAR TRANSFORMATION OF THE THETA-FUNCTIONS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XXI. (1886), pp. 77–84.]

THE notation is that of Smith's* "Memoir on the Theta and Omega Functions," differing from Hermite's only in a factor, $\Im_{m,n}$ (Smith) $= 2^{n/4}\Im_{m,n}$ (Hermite); and it is in consequence of this that the factor $2^{n/4}$ occurs in the expression presently given for Λ. Hermite's Memoir "Sur quelques formules relatives à la transformation des fonctions elliptiques" appeared in *Liouville*, t. III. (1858), pp. 27–36.

Writing

$$\omega = \frac{a+d\Omega}{c+d\Omega},$$

where $ad-bc=1$, the formula of transformation is

$$\Im_{m,n}\left[(a+b\Omega)\frac{m^2}{2},\Omega\right] = C\exp\left(-i\pi(a+b\Omega)\frac{m^2}{2}\right)\Im_{m,n'}\left(\frac{m\cdot a}{2},\omega\right);$$

where:

$$\begin{aligned} m' &= cu + b\cdot n, \\ n' &= cm + a\cdot n, \end{aligned}$$

or:

$$\begin{aligned} m' &= cm + b\cdot n, \\ n' &= cm + a\cdot n. \end{aligned}$$

We have, according as b is positive or negative,

$$C = \exp\left[-\frac{i\pi}{\sqrt{(-(c+d\Omega))}}\right] = \exp\left[-\frac{i\pi}{\sqrt{(c+d\Omega)}}\right],$$

where for each case the square root is to be taken in such wise that the real part is positive; Λ, H are eighth roots of unity; viz. in each case

$$\Lambda = \exp\left[-\frac{1}{2}i\pi(\alpha m^2 + 2bmn + dn^2 + 2\alpha\beta n + 4\beta\alpha n + ab\gamma\delta)\right]e^{m\sqrt{(-1)/2}},$$

*[H. J. S. Smith, *Collected Mathematical Papers*, vol. II., pp. 415–634.]

and, according as b is positive or negative,

$$H = \left(\frac{b}{a}\right) e^{i\pi a}, \qquad a \text{ odd}, \qquad H = \left(\frac{-b}{a}\right) e^{i\pi(a-1+3b+\beta)}, \qquad a \text{ odd},$$

$$H = \left(\frac{a}{b}\right) e^{i\pi(-b+2-a-1)/2}, \quad b \text{ odd}, \qquad H = \left(\frac{-a}{-b}\right) e^{i\pi(a-1+3b+\beta-b+2)/2}, \quad b \text{ odd};$$

where $\left(\frac{a}{b}\right)$, &c., are the Legendrian symbols as generalised by Jacobi; if a and b are each of them odd, the formula for the case of a even or odd and b odd respectively are equivalent to each other, and either may be used indifferently.

To fix the ideas, I assume throughout that b is positive and odd; the proper formulae then are

$$C = \frac{H}{\sqrt{(c+d\Omega)}};$$

Λ as above, and

$$H = \left(\frac{a}{b}\right) e^{i\pi(-b+2-a-1)/2}.$$

I will also, for greater convenience, assume that c is odd; in consequently even, viz. the numbers a and c are odd and those even, or else a and b are odd and the other even.

The equation $\omega = \frac{a+d\Omega}{c+d\Omega}$ gives $\Omega = \frac{-a+c\omega}{-d+b\omega}$, and thence

$$a + b\Omega = -\frac{1}{-d+b\omega}.$$

Comparing the expressions for ω, Ω, we observe that we may interchange ω and Ω, writing also $-d$, b, c, $-a$ for a, b, c, d; and changing the other letters, we may for u, b, β, m, n, m', n', write

$$\frac{u}{c+d\Omega}, \quad b, \quad a, \quad m', \quad n', \quad m, \quad n,$$

where

$$m' = -dm - bn, \quad n' = -cm - an;$$

the formula then becomes

$$\Im_{m',n'}\left[(-d+b\omega)\frac{m\cdot}{2}, \omega\right] = C' \exp\left[-i\pi(-d+b\omega)\frac{u^2}{2}\right] \Im_{m,n}\left(\frac{u}{c+d\Omega}, \Omega\right),$$

or, as it is the same thing,

$$\Im_{m',n'}\left[(a+b\Omega)\frac{m\cdot}{2}, \omega\right] = C' \exp\left[-i\pi(a+b\Omega)\frac{u^2}{2}\right] \Im_{m,n}\left(\frac{u}{c+d\Omega}, \Omega\right);$$

or, observing that the left-hand side is $= (-1)^{mn'} \Im_{m,n'} \left[\frac{u}{2}, \omega\right]$, we have

$$\Im_{m,n'} \left[(a + b\Omega)\frac{m\cdot}{2}, \omega\right] = \frac{C'}{(-1)^{mn'}} \exp\left[-i\pi(a + b\Omega)\frac{u^2}{2}\right] \Im_{m,n}\left(\frac{u}{c + d\Omega}, \Omega\right),$$

an equation which is of the same form as the original equation of transformation, and is to be identified with it.

We have

$$\begin{aligned} m' &= -dm - bn = -dm + b\Omega \cdot n, \quad = -m + bd(-n + cm), \\ &\quad + b \cdot bd(-n + cm) = -bd, \\ n' &= -cm - an = -(cm + an) = m \cdot bd(-n + cm) = -bn(-n + cm) - 1, \end{aligned}$$

values which may be written

$$m' = -dm - bn + Pa = -bd, \quad m \cdot b \cdot bd(-n + cm) - 1,$$

or, since $\Omega - d - b\omega = -bd$,

$$m' = -d + bd \cdot (-n + cm) = b\omega \cdot m = -d \cdot \quad m = cm + b \cdot n - 1,$$

$$n' = -c \cdot \quad m \cdot \quad a \cdot \quad n - 1, \qquad n' = c \cdot \quad m + a \cdot \quad n - 1;$$

P and Q being integers. In fact, P is an integer if b is odd or if bn is even, and similarly for Q.

The left-hand side is $\Im_{m',n'} \left[(a + b\Omega)\frac{u}{2}, \Omega\right]$, which is in virtue of P even, and which depends only on the values of m and n.

We have

$$CC' = (-1)^{mn' - mn + m'n' - m'n},$$

which is

$$= (-1)^{mn - m'n'}$$

this condition will be considered in the sequel.

The formula in question is in fact identifiable to that into which it transforms when, observing that the left-hand side is unchanged by the change of u, b, c, d, m, n, m', n', into $-d$, b, c, $-a$, m', n', m, n;

$$CC' = (-1)^{mn - m'n}, \quad CC' = (-1)^{mn' - m'n}.$$

We have

$$CC' = \frac{HH'}{\sqrt{(c + d\Omega)} \cdot \sqrt{(-(c + d\Omega))}},$$

where the square roots are taken in such wise that the real part is positive; hence

$$CC' = \frac{HH'}{i\sqrt{(c^2 + 1)}} \quad \text{or} \quad \frac{HH'}{i\sqrt{(-(c^2 + 1)d)}},$$

according as c is even or odd; for, taking the radius vector for positive, we have that the square roots of $c + d\omega$ and $\sqrt{(-(c + d\omega))}$ are of the form

$$\sqrt{(c + d\omega)}, \quad \frac{1}{i}\sqrt{(c + d\omega)} \quad \text{or} \quad \frac{1}{i}\sqrt{(-(c + d\omega))}, \quad \sqrt{(-(c + d\omega))} = \sqrt{(c + d\omega)},$$

where the square roots taken in such wise that the real part is in each case positive; 43 − 2 where $\sqrt{(c+1)}$ means

$$\sqrt{(-(c+1)d)}\sqrt{(-)}\cdot\sqrt{(-(c+d)\Omega)};$$

the last-mentioned two square roots being so taken as just explained; and we have therefore

$$\Lambda = \exp\left[-\tfrac{1}{2}i\pi(\alpha m^2 + 2bmn + dn^2 + 2\alpha\beta n + 4\beta\alpha n + ab\gamma\delta)\right] e^{m\sqrt{(-1)/2}},$$
$$\Lambda' = \exp\left[-\tfrac{1}{2}i\pi(-dn^2 + 2bm'n' - am'^2 - 2\alpha\beta m' - 4\beta\alpha m' - bd\beta)\right] e^{m\sqrt{(-1)/2}},$$

viz. the values of Λ obtained from the value of Λ by the change of a, b, c, d, m, n, m', n', into $-d$, b, c, $-a$, m', n', m, n; and writing

$$A = \exp\left[-\tfrac{1}{2}i\pi\{-am^2 + 2bmn + dn^2 + dn'^2 - 2bn'm' + am'^2 + 2\alpha\beta(n-m') + 4\beta\alpha(m-n') - ab\gamma\delta - b$$

the last-mentioned two square roots being so taken as just explained, and we have

Representing the second and third of the last-mentioned equations by

$$4 = \exp\left(-i\pi\rho^{-mn'-m'n}\right),$$

the last-mentioned two squares roots being also explained; and we have to express the term divisible by 4,

$$\rho = -mn'(2 + ab\gamma\delta + bd\beta).$$

To calculate $\Delta + \Delta'$, we have

$$dm - bn = m + bd\cdot(n - c\cdot m), \quad m = -d\cdot m - b\cdot n - 1,$$
$$-cm + an = -n - ac\cdot(n - c\cdot m), \quad n = c\cdot m + a\cdot n - 1,$$

and thence

$$-2dm^2 + 2bn^2 = -mn' + ac\cdot n' - bd\cdot m'^2 + abcd^2 - 2bd(-n'2 + cdn) - 2acd(2 - cd) = -2bdmn + \&c.$$
$$(-am + bn)^2 = -mn'(-2bd - bnn' + 2abcd) - 2bd(-n' + cn) + 2acd(-a' + cn) - 2bdcd(-n' + cn) =$$
$$- -2acd(-a' + cn) - bd\cdot 2a^2 + 2abdn - 2acd(-n' + ca) + 2abcd^2 = 4acdn,$$

consequently

$$-adn^2 + 2bnn' + 2bcmn' - 2bdmn + 2acmn + bdn^2 + 2bdcmn = 2acdn(n + 2cdn) - 2bdcmn = 4cdmn,$$

also

$$-2bdmn = -2cdbn^2 - 2bd^2n^2 + 2acd\cdot n + 2bcd\cdot n^2 + 2bd\cdot c\cdot 2bd\cdot 2bd\cdot bd,$$
$$-2bn' = -2adcb - 2cdb\cdot 2bd\cdot 2bn^2 + 2cdb\cdot 2cd\cdot 2bd\cdot bd\cdot 2bcb,$$
$$-dn'^2 + dn'^2 = -2bcd^2\cdot dcb - 2dcb\cdot dn + 2cdn^2 + 2bcd\cdot dcb\cdot bcb + 2cdb\cdot bd\cdot bd,$$

whence, adding, we obtain

$$\Delta + \Delta' = -dn^2 - 2bnn' + 2bcmn' - dn^2 + 2cdmn' - 2bcdmn' - bcdm + 2bcd\cdot n^2 + 2cd\cdot dcb + 2cd\cdot bd\cdot bd,$$

and, adding to this the expression of Δ, we find

$$\Delta + \Delta' = -2bnn' + 2bnn' - 2bcmn' + 2cd\beta - 2dnn' + 2dcn' - 2bcd - bdcd + 2bd\cdot bd,$$

The coefficient of u is $= 2bd(-n+cm) - 4cd + ad(-c\cdot m^2 + a\cdot n) + 2(b-1)(c\cdot n)$. Hence writing

$$m\cdot n' = 2bd(-n+cm) - ad(-m\cdot n^2 + a\cdot n) + 2bnn(-1)(c\cdot n - n),$$

observe that $a\Omega n - 1$, observing in every case $\Omega \cdot mod$, writing also $4bn \cdot 2cdn$, and consequently

$$4bn\cdot n' = ad(\beta m - n)(-\beta + b) + (-\beta + b)(b-1)(c+d)(b-1)(n-2m).$$

where the even values of n may be omitted. We have moreover

$$mn - m'n' = -bd(-n+cm)\beta - ab(-m\cdot n^2 + a\cdot n)\beta + (-\beta + b)(b-1)bd,$$

and these, since β is even, may be neglected, and the result is simply

$$mn - m'n' = -bd\beta - ad\delta\beta,$$

and thus

$$mn - m'n' = ad\delta\beta + bd(-2+cd)\beta - 2bn\delta\delta(-a+cd)\beta,$$

we have also

$$m - n' = -d\cdot m - bd - (c\cdot m - a\cdot n + 1) = -(c+d)(b)\beta + 1,$$

$$n - m' = -m + d\cdot m + ba = ad(b-n).$$

If even of which disappear, leaving the result, $u\sqrt{(-1)/2}$ becomes equal to $4ms\Omega$.

The next thing is to express the term divisible by 4: there is at once a term given by $4(b-1)(c\cdot 2bn)$, and so we may write therefore

$$4(b-1)(c\cdot 2bn) + 4(-bd)(2+cd) + (-bd)(b-1)(c\cdot 2cdb) = 0,$$

via. the values of Δ is given by Abel's theorem, then in particular each of the four equations becomes

$$4(b-1)(c\cdot d\cdot n) = -2bdcn,$$

where, this is positive value of β is positive; and thus

$$mn - m'n' = -dn\beta + bd(c-1)(c\beta - 1),$$

the equation which was to be verified.

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ON THE THEORY OF SEMINVARIANTS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XXI. (1886), pp. 1–17, [13, 14, 16].]

IN my "Mémoire sur les Hyperdéterminants," *Crelle*, t. XXX. (1846), pp. 1–37, [13, 14, 16],* I gave an investigation for the number of and relations between the quartic invariants of a given binary quantic, or what is the same thing to the cubic seminvariants of a given weight, and I propose at present (considering the formulae in this point of view) to further develop the theory in regard to the solution of the systems of linear equations obtained in the manner in question. But I first commence with a remark: I recall that the notation is the ordinary hyperdeterminant notation: for instance,

$$U = (a, b, c, d)(x, y)^3, \quad V = (a, b, c, d)(x, y)^3,$$

$$\begin{aligned} 12VV \cdot 1 + 2, \quad VVV \quad &= D_{a,a'} VVV, \\ 12VV \cdot 12 &= D_{a,a'} D_{b,b'} VVV, \\ \&\text{c.} \quad &= D_{a,a'} D_{b,b'} \cdot D_{c,c'} VVV, \end{aligned}$$

and that, when the variables $(x, y), (a, b), (a', b')$ are ultimately replaced by (x, y), then these forms denote, each of them, an invariant.

The investigation* is as follows:

$$\begin{aligned} UU(12) &= (a, b, c, d)(b', a)(b', d') \cdot -2(ad' + a'd - bc' - b'c) + 9(bc' + b'c), \\ &= 12P(a' \cdot b'^2 - 5abb' \cdot b'^2 \cdot 3a^2bb' + b'^2 \cdot a'd' - 5b'd \\ &\quad + a \cdot a' \cdot -25, \\ &\quad + 2(cc' - d^2), \end{aligned}$$

and that, when the variables $(x, y), (a, b), (a', b')$ are ultimately replaced by (x, y), then these forms denote, each of them, an invariant.

The investigation is as follows:

$$\begin{aligned} UVUV &= UUVV - U_xV \cdot VWV \cdot UWV, \\ UUVW &= UWUV \cdot VWWV, \\ &= D_{a,b'} VVWV = D_{b,a'} VWVV, \\ &\&\text{c.} \end{aligned}$$

*[This Collection, vol. I., p. 100; see also, vol. I., p. 157.]

Writing for shortness

$$12, 34 = \mathfrak{A}, \quad 13, 42 = \mathfrak{B}, \quad 14, 23 = \mathfrak{C},$$

we have

$$D_{a,d}\mathfrak{A} + D_{b,c'}\mathfrak{B} + D_{c,b'}\mathfrak{C} = 0.$$

Suppose $U = V = W = X$; we have the derivatives which correspond to the same value of δ, w, we have from these

$$D_{a,d}\mathfrak{A}, \quad D_{b,c'}\mathfrak{B}, \quad D_{c,b'}\mathfrak{C} = 0, \quad D_{d,a'}\mathfrak{D}_{x'}\mathfrak{A} = 0,$$

in the language of Prof. Sylvester, $\alpha_{i,k}$, $b_{i,k}$, &c., are here scalars. The invariant is

$$(\alpha\beta_{i,b}b_c - b_{i,b}b_{c,d})\mathfrak{A} = \mathfrak{B}_{i,k}b_{c,d}\mathfrak{C} = b_{i,b}b_{c,b'},$$

$= an^2 - 3b^2 + 2c^2, =$ &c., are here equal; and the invariant is

$$= 2(\alpha b - 4),$$

and so in other cases. To apply this directly to the theory of seminvariants, observe that, in the language of Sylvester's $\alpha_{i,b}$, $b_{i,b}$, $b_{i,b}$, $b_{i,b}$, &c., are here scalars. The invariant is

$$(ax)_2(ax)_3(ax)_4 = D_{a,d}D_{a,c}D_{a,b}V \cdot V = 4ax,$$

that is,

$$\alpha = 4f - 3g + 2h,$$

and so in other cases. To apply this directly to the theory of seminvariants, observe that the Cliebschian notation, for instance

$$(\alpha, \beta, c, d)(x, y)^3 + 4(x + b)(x + d)y + 6((c + b)y + 4(c + d)y + 4y) = 4ax,$$

viz. in the language of Prof. Sylvester, $\alpha_{i,b}$, $b_{i,b}$, &c., are here scalars. The invariant is

$$(\alpha\beta_{i,b}b_c - b_{i,b}b_{c,d})\mathfrak{A} = \mathfrak{B}_{i,k}b_{c,d}\mathfrak{C} = b_{i,b}b_{c,b'},$$

$= an^2 - 3b^2 + 2c^2, =$ &c., are here equal; and the invariant is

$$= 2(\alpha b - 4),$$

and so in other cases. To apply this directly to the theory of seminvariants, observe that, in the Cliebschian notation, for instance:

$$(\alpha\beta + \alpha\delta - \beta\gamma)\mathfrak{A} = 4(\alpha\delta - \beta\gamma),$$

that is,

$$a = bf - 3g + 2h,$$

$a = bf - 3g + 2h$, $\alpha\beta = 4f - 3g + 2h$,

$$a = bf - 3g + 2h,$$

a seminvariant; the

$$(a-\beta)^2(\alpha-\gamma) = -2\alpha\beta + a\gamma + 4a(c-\beta) - 8c\delta + a\gamma\beta - 3\beta + 4ac\delta,$$

$\beta = 4f - 3g + 2h$, &c., are scalars: writing each of f, g, h for the weight). We have, as before,

$$\mathfrak{A} + \mathfrak{B} + \mathfrak{C} = 0,$$

and we again see that the theory is one of the number of relations between the cubic seminvariants of a given weight. Observe also that, for each of the highest powers of a, the cubic seminvariants $\mathfrak{CVC}$, $\mathfrak{CVC}$, are the differences $D_{a,a}\mathfrak{A}$, &c., but in many cases the seminvariants $D_{a,a}\mathfrak{A}$, &c., are ultimately replaced by (x, y), then these forms denote, each of them, an invariant.

Or, we may use the Cliebschian notation, for instance, observe that

$$(\alpha, \beta, c, d)(x, y)^3, \quad \alpha = -cu + bv = av + bw, \quad \text{\&c.},$$

that is,

$$a = c - b = -a,$$

via. in the language of Prof. Sylvester, $\alpha_{i,b}$, $b_{i,b}$, &c., are here scalars. The invariant is

$$(\alpha-\beta)^2 = a + (\alpha+\beta)^2,$$

$$= 4(\alpha - 2b + b^2),$$

where α, β, … are scalars.

The invariant is

$$(\alpha-\beta)^2(\alpha-\gamma) = \alpha^2 + \alpha\beta\alpha + 2\beta\gamma\alpha\beta + \beta^2,$$

$\alpha = -\beta$, $\beta = -\alpha$, $\alpha = -\beta$, $\beta = -\alpha$,

$$\alpha = -\beta\beta\beta\beta\beta - \beta = 0.$$

Hence

$$\mathfrak{A} \cdot \mathfrak{V} = 0.$$

We may also for independent derivatives, when f is even, the terms of the series $D_{a,a+1}$, $D_{a+1,b+1}$, ..., and when f is odd, those of the series $D_{a+1,a+2}$, $D_{a+2,b+2}$, ..., continuing each series until the last term in which $D_{a+i,b+i}$ would be equal to zero in virtue of the value of δ, w (notice that the values of δ, w must be in such relation, so that there is no need of a sequence of the seminvariants $D_{i,m}$ are independent the series of the order term $D_{a,a+1}$); the like considerations apply to the other derivatives $D_{a,a+1}$, &c., but inasmuch as we must keep at least a of order a different, the same set must necessarily be in use, but the same set of independent seminvariants will of course be obtained. The number of these is also f, only the same set will be the same as we are considering; the number of independent seminvariants is in fact equal to the number of δ, w, &c., which is independent of δ, w, &c., in the form $\mathfrak{V}_{a,b+m}, \mathfrak{V}_{a,b+m}, \ldots$. We may say that all these seminvariants of the highest powers of δ, w are mutually independent, and as such are connected by linear relations between them, the coefficients of these depending on the seminvariants of the lowest powers of δ, w; and as such these linear relations are linear functions of the seminvariants, the connecting relations being also linear functions; the seminvariants of the highest powers are also linear functions of those of the lowest, the seminvariants of the highest powers being also represented by the corresponding numbers in the form $\mathfrak{V}_{a,b+m}, \mathfrak{V}_{a,b+m}, \ldots$. We may say that all this means that of any independent seminvariants of the form $\mathfrak{V}_{a,b+m}, \mathfrak{V}_{a,b+m}, \ldots$, we have any independent linear relations between the corresponding numbers in the form $\mathfrak{V}_{a,b+m}, \mathfrak{V}_{a,b+m}, \ldots$.

Or, we may say that the number of independent linear relations between the seminvariants of the highest powers is, in proportion to the number of independent seminvariants of the lowest powers, &c., of the form $\mathfrak{V}_{a,b+m}, \mathfrak{V}_{a,b+m}, \ldots$, equal to the number of independent linear relations between the seminvariants of the lowest powers, &c., of the form $\mathfrak{V}_{a,b+m}, \mathfrak{V}_{a,b+m}, \ldots$.

We will presently give the solution of the under-mentioned system; but beginning with the second and weight w as we have here, the number of solutions will be the following table:

	$D_{a,d}$	$D_{a,d+1}$	$D_{a,d+2}$	$D_{a,d+3}$	
$\mathfrak{D}_{ad}$	D_a	0,	D_a	$\frac{1}{2}D_a$	0,
$\mathfrak{D}_{ad+1}$	$-D_a$	$\frac{1}{2}D_a$	0,	$\frac{1}{2}D_a$	0,
D_{ad}	0,	0,	D_a	$-D_a$	$\frac{1}{2}D_a$
D_a	0,	0,	0,	D_b	$-D_b$
$\mathfrak{D}_{ad+1}$	0,	0,	0,	0,	
D_{ad+1}	0,		0,	0,	$-D_a$

We may use generally with the solution of the equations of column corresponding numbers

$$p, \quad q, \quad p+1, \quad p+1, \quad p+1, \quad \ldots,$$

there is for example a single linear relation between the orders of the relations $5, 7, 8, 9, 10$ in each of the orders $5, 8, 9, 10$ on the form $\mathfrak{V}_{a,b+m}, \mathfrak{V}_{a,b+m}, \ldots$, on the relations between the orders of $D_a = \frac{1}{16}D_b$. The above considerations were given as follows:

For any value of f except $f = 2$, 3, or 4, the table commences, for f even and f odd respectively, in the following manner:

	$D_{a,b+1}$	$D_{a,b+2}$	$D_{a+1,b+2}$	$D_{a+1,b+3}$	$D_{a+2,b+3}$						
$\mathfrak{D}_{a,b+1}$	$\frac{1}{2}f$	0									
$\mathfrak{D}_{a+1,b+1}$	$-\frac{1}{2}f$	$\frac{1}{2}f$	0								
$\mathfrak{D}_{a+1,b+2}$	0	$-\frac{1}{2}(f-1)$	$\frac{1}{2}(f+1)$	0							
$\mathfrak{D}_{a+2,b+2}$	0	0	$-\frac{1}{2}(f-2)$	$\frac{1}{2}(f+2)$	0						
$\mathfrak{D}_{a+2,b+3}$	0	0	0	$-\frac{1}{2}(f-3)$	$\frac{1}{2}(f+3)$	0					
$\mathfrak{D}_{a+3,b+3}$	0	0	0	0	$-\frac{1}{2}(f-4)$	$\frac{1}{2}(f+4)$					

but beyond this I do not know the law of the series.

Before going further, I remark that if $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, instead of the foregoing values, denote respectively,

$$\begin{aligned}
\mathfrak{A} &= (\alpha\delta - \beta\gamma)(\alpha'\delta' - \beta'\gamma'), \quad = (\alpha\beta'_2 + \alpha\beta')(\gamma\delta'_2 - \delta\gamma'),\\
\mathfrak{B} &= (\alpha\delta + \beta\gamma)(\alpha'\delta' + \beta'\gamma'), \quad = (\alpha\beta'_2 - \alpha\beta')(\gamma\delta'_2 + \delta\gamma'),\\
\mathfrak{C} &= (\alpha\beta + \alpha'\beta')\Lambda, \quad = (\alpha\delta + \beta\delta')(\alpha\gamma + \beta\gamma' - \delta\delta'),
\end{aligned}$$

then equations

$$\mathfrak{A} + \mathfrak{B} + \mathfrak{C} = 0,$$

and the same theory applies to the cubic derivatives $\mathfrak{D}_{a,b+m}\mathfrak{V}^{2/m}\mathfrak{C}\mathfrak{V}\mathfrak{V}$, that is, in the like relations between the coefficients of the highest powers of the cubic derivatives.

Or, we may apply this directly to the theory of seminvariants. We take then $(x, y)^4$ to denote the binary quartic, and we have for instance

$$(\alpha, b, c, d, e)(x, y)^4 = au^4 + 4bu^3v + 6cu^2v^2 + 4duv^3 + ev^4 = (\alpha, b, c, d, e),$$

that is,

$$\alpha = a, \quad b = -b, \quad c = c, \quad d = -d, \quad e = e,$$

a seminvariant; the like relations between the coefficients of the highest powers of the cubic derivatives, $\mathfrak{D}_{a,b+m}\mathfrak{V}^{2/m}\mathfrak{C}\mathfrak{V}\mathfrak{V}$. We have

$$\mathfrak{B} = (\alpha - \beta)(\alpha\gamma - \beta\delta) = -2\beta\alpha + a\gamma + 4a(c - \beta) - 8c\delta + 4ce - 3\beta + 4\delta\gamma,$$

a seminvariant.

The like is the case as in other cases. Writing here

$$\mathfrak{A} + \mathfrak{B} + \mathfrak{C} = 0,$$

and we see that the question relates also to the cubic seminvariants of a given weight. Observe also that, for each of the highest powers of δ, w, $\mathfrak{C}\mathfrak{V}\mathfrak{C}$, $\mathfrak{C}\mathfrak{V}\mathfrak{C}$ are the differences $D_{a,a}\mathfrak{A}$, &c., but the same set of independent seminvariants are also represented by $(\alpha-\beta)^2$, &c..

Suppose that a is odd, and, in particular, $D_{a,d+m}\mathfrak{V}_{a,d}\mathfrak{V}\mathfrak{V}$, &c. that is,

$$D_{a,d+m} = -D_{a,d}\mathfrak{V}\mathfrak{V}, \quad D_{a,d+m} = -D_{a,d}\mathfrak{V}\mathfrak{V},$$

that is,

$$\mathfrak{V}_{a,b+m}\mathfrak{V}_{a,b+m} = 0, \quad \mathfrak{V}_{a,b+m}\mathfrak{V}_{a,b+m} = 0;$$

Hence

$$\mathfrak{A} = 0;$$

or, in like manner if a cubic seminvariant, that is,

$$(b, b, b)(\delta, x, y)^3 = -bv^3 + 4bv^2y + 4(\alpha + by)y^2 + (a + by)y^3,$$

that is,

$$\alpha = -a, \quad \beta = -b, \quad \delta = -d, \quad \gamma = -c, \quad \delta = -e,$$

a seminvariant. Writing here

$$\mathfrak{A} + \mathfrak{B} + \mathfrak{C} = 0,$$

and we see that the question relates to the number of relations between the cubic seminvariants of a given weight. Observe also that, for each of the highest powers of δ, w, the cubic seminvariants $\mathfrak{CVC}$, $\mathfrak{CVC}$ are the differences $D_{a,a}\mathfrak{A}$, &c..

The like remarks would apply for f odd, in the case a a quartic seminvariant, the like remarks apply also, if instead of the seminvariants f odd, we have the seminvariants f even; we have here a similar set of seminvariants $D_{a,d}\mathfrak{V}$, $D_{a,d+m}\mathfrak{V}$, &c., and the seminvariants $D_{a,d}\mathfrak{V}$, &c. are also as above; in many cases the seminvariants are not all of them simply equal, but coefficients in the relations A, B, C, &c., respectively, I call them

$$A, \quad B, \quad C, \quad D, \quad E, \quad F, \quad G, \quad H, \quad I, \quad J, \quad K, \quad L, \quad M,$$

Suppose first α is odd, the in the first

$$D_{a,d}\mathfrak{VV} = 1, \quad D_{a,d+m}\mathfrak{VV} = 0,$$

and similarly $C + 2D + E = 0$, &c.; that is, we have

$$\begin{aligned} A &= 0, \\ B + C &= 0, \\ C + 2D + E &= 0, \\ D + 3E + 3F + G &= 0, \\ &\text{\&c.} \end{aligned}$$

The resultant way of solving them is to express the other functions in terms of B, D, F, H, &c., viz. we thus have

	A	B	C	D	E	F	G	H	I	J	K	L	M
X	0	1	−1	0	1	0	−10	0	+135	0	−2073		
D			1	−2	0	3	−15	0	355	0	−5410		
F				1	−3	0	6	−126	0	1693			
H					1	−4	0	10	0	−150	0	−396	
J						1	−5	0	15				
L								1	−6				

read according to the columns

$$
\begin{aligned}
A &= 0, \\
B &= B, \\
C &= -B, \\
D &= D, \\
E &= B - 2D, \\
F &= F, \\
G &= -3B + 3D - 3F, \\
&\text{\&c.}
\end{aligned}
$$

and, by way of verification of the numbers, observe that the sum of the numbers in a column is 1 and -1 alternately.

The formulae are true for any odd value of α whatever; but they require an explanation. On the right hand value of α, &c., are not all of them advisory, but they are also of the same powers of δ, w. The proper sequence ought to be obtained, but it is not derivable from these by Captain MacMahon's theorem, viz. the number of independent seminvariants is in fact equal to the number of δ, w, of the non-unitary symmetric functions $\mathfrak{V}\mathfrak{V}$ of the proper weight: for δ even, the forms are 2222 and 3322; so that there should be only two arbitrary terms.

moreover

$$B + C = w^{-1}(\mathfrak{B} + \mathfrak{C}) = 0,$$

and similarly $C + 2D + E = 0$, &c.; that is, we have

$$\begin{aligned} A &= 0, \\ B + C &= 0, \\ C + 2D + E &= 0, \\ D + 3E + 3F + G &= 0, \\ &\&c. \end{aligned}$$

The resultant way of solving these is to express the other functions in terms of B, D, F, H, J, L alternately.

The formula are true for any odd value of f whatever; but they require an explanation. The arbitrary terms B, D, &c., are not all of them arbitrary; but they are also seminvariants of the same weight, and therefore connected by linear relations. The proper sequence ought to be obtained, but it is not derivable from these by Captain MacMahon's theorem, viz. the number of independent seminvariants is in fact equal to the number, δ, w, of the non-unitary symmetric functions $\mathfrak{V}\mathfrak{V}$ of the proper weight: for δ even, the forms are 2222 and 3322; so that there should be only two arbitrary terms.

We may have, for any odd value of f, writing for shortness 900, 810, &c., instead of D_{900}, D_{810}, &c.,

$$\begin{aligned}
900 &= 0,\\
810 &= B,\\
720 &= -B,\\
630 &= D,\\
540 &= B - 2D,\\
450 &= F,\\
360 &= -3B + 3D - 3F,\\
270 &= H,\\
180 &= 12B - 2D + 14F - 4H,\\
090 &= J,
\end{aligned}$$

But we have $900 = 0$, $810 + 720 = 0$, $720 + 270 = 0$, $630 + 350 + 450 = 0$, $540 + 350 = 0$; that is,

$$\begin{aligned}
900 &= 0,\\
810 &= B,\\
720 &= -B,\\
630 &= D,\\
540 &= B - 2D,\\
450 &= F,\\
L &= 0,
\end{aligned}$$

all satisfied by $F = -B + 2D$, $H = B - D$, $L = 0$, the proper answer to stop at the equation for 540, the system is then

$$\begin{aligned}
900 &= 0,\\
810 &= B,\\
720 &= -B,\\
630 &= D,\\
540 &= B - 2D,
\end{aligned}$$

equations which may be considered as expressing the several functions 900, 810, 720, 630, 540, 450, &c., in terms of $B = 810$ and $D = 630$. They agree, as they should do, with a foregoing result,

$$\begin{aligned}
900 &= 0,\\
810 &= B,\\
720 &= -B,\\
630 &= D,\\
540 &= B - 2D,
\end{aligned}$$

and it would of course be possible to express the remaining forms $D_{a,b}$, $D_{a,c}$, &c., 432, and 333 in terms of $B = 810$ and $D = 630$. But observe that the speciality of